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PDE-based containment control of linear multi-agent systems

Joachim Deutscher¹ | Nick Jung

Institute of Measurement, Control and Microtechnology, Ulm University, Ulm, Germany

Correspondence

Joachim Deutscher, Institute of Measurement, Control and Microtechnology, Ulm University, Ulm, Germany.

Email: joachim.deutscher@uni-ulm.de

[Correction added on 22 January 2025, after first online publication: the copyright line was changed.]

Abstract

This contribution considers the containment control problem for linear multi-agent systems (MAS) using continuum models in form of linear parabolic PDEs. For this, a networked controller is designed, that ensures the asymptotic convergence of finite-dimensional agents into a containment area specified by dynamic leaders. This problem is traced back to an output regulation problem for a continuum model independent of the number of agents, which also takes disturbances into account. A prescribed formation is assigned for the MAS in the containment area by using Bézier curves. Using bilateral backstepping a tracking controller is designed, which allows to distribute the control effort between the two boundary agents. The states are estimated by an infinite-dimensional disturbance observer. For this, both a folding observer with network communication and two boundary observers without network communication are introduced. A fully distributed solution of the containment control problem is obtained by employing a continuum signal model observer, which communicates the states of the leader and disturbance model to all agents. The desired communication topology is imposed after the design by a spatial discretization. A simulation example illustrates the effectiveness of the new method for the networked controller design.

KEYWORDS

bilateral backstepping, containment control, continuum models, multi-agent systems, parabolic systems

1 | INTRODUCTION

The networked control of *multi-agent systems* (MAS) is still a very active research area in the control community. This is due to the large variety of control problems, which lead to this system class. Examples range from formation control of UAVs, robots or vehicles to synchronization problems in electrical networks including micro grids (see the recent monographs^{1,2} for applications). While a great deal of networked control methods use finite-dimensional control theory (see, e.g., References 1–3 for an overview), the application of control theory for PDEs to networked control by making use of continuum models is a relatively new research topic. This method has the inherent advantages that the networked controller design becomes independent from the number of agents and the communication topology. Different from finite-dimensional methods, this approach allows to freely specify the synchronization dynamics, which for ODEs

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is restricted due to the appearing simultaneous stabilization problem. Furthermore, a desired communication topology can be imposed by a suitable spatial discretization after the controller design. One research direction is to apply backstepping methods (see, e.g., Reference 4 for an overview) to the continuum model stabilizing prescribed steady states, which correspond to desired agent formations (see References 5 and 6). Refinements of this approach are considered in Reference 4, where flatness-based methods are utilized to achieve a finite-time agent deployment. Another research direction is to apply Lyapunov-based methods to deploy agents on prescribed curves (see, e.g., References 7 and 8).

An interesting cooperative control problem is *containment control*. This amounts to designing a networked controller, which ensures that all agents converge to a containment area specified by the convex hull of the leaders. Then, the leaders are able to guide the agents through, for example, a hazardous area. Potential applications of the problem are search and rescue, cooperative transport and handling of hazardous material (see Reference 3 (Chapter 5)). The leaders could be agents equipped with the necessary sensors to detect hazardous obstacles like holes in the ground or mines. This way the leaders can determine a safety area (the containment area) and guide the followers securely through the hazardous area. In the literature, only finite-dimensional methods are used to solve the containment control problem for the ODE-Agents directly. In the context of output regulation, cooperative signal model observers are designed, which achieve containment through feedforward control (see, e.g., Reference 9). Alternatively, the design can also be performed using the internal model principle, which, in contrast to the first approach, is also robust to model uncertainties (see, e.g., Reference 10). An overview of other methods for solving the containment control problem using ODE methods can be found in Reference 11. Common to all available solutions in the literature is that the desired formation in the containment area is dictated by the network topology and the agents initial conditions, which means the formation can't be specified although this is of interest for many applications. So far the containment control problem has not been considered in the framework of continuum models, allowing to make use of the corresponding advantages. It has to be noted that none of the methods presented in References 5–8,12 using continuum models are able to solve the containment control problem with prescribed formations in the containment area.

This article considers the solution of the containment control problem for MAS using a continuum model described by a parabolic PDE. The networked controller design is traced back to an output regulation problem for a bilaterally actuated parabolic PDE with dynamic boundary conditions. A main result of the article is a suitable choice of the reference trajectory for the agent continuum. In particular, it is shown that *Bézier curves* can be utilized to ensure a systematic solution of the containment problem. These types of curves are fundamental in computer graphics and are actually used in most computer graphic programs (see Reference 13). It is shown in the article that it is possible to deploy the agents on formations described by Bézier curves in a 2D-containment area such that containment is ensured, allowing to consider a rich class of agent formations. The resulting feedforward control problem is solved by taking disturbances into account that are described by the solution of ODEs. Then, the leaders, also modeled by ODEs, are able to guide the prespecified agent formation through a desired path in the plane. In order to ensure stable tracking error dynamics, the recent results^{14,15} for bilateral backstepping are extended to deal with parabolic PDEs with dynamic boundary conditions. This allows to systematically exploit the degrees of freedom contained in the bilateral actuated PDE and allows to assign a desired convergence rate for the agent deployment. As the agents and the leaders positions as well as the disturbance model states are needed for the feedforward and the state feedback two kinds of observers are proposed. The first observer uses a folding approach, where the agent with absolute and relative position measurement can be chosen freely. This, however, requires the communication of the measurements to the boundary agents. The latter drawback can be avoided with the second approach, which consists of two boundary observers using boundary agents absolute and relative position measurements. In both cases, the estimated disturbance model states have to be communicated to the remaining agents. To this end, a continuum formulation of a cooperative signal observer proposed in Reference 16 is presented for the first time and a systematic design procedure is developed. Consequently, the agents only need to know the basic shape of the agent formation, but get the information of the actual Bézier curve lying in the containment area through communication. This results in a new, systematic and fully distributed approach to guide prespecified agents formations through the plane by means of containment control.

The next section introduces the considered containment control problem, which is followed by the continuum formulation in Section 3. Then, the feedforward controller is presented in Section 4 and the design of the bilateral tracking controller is presented in Section 5. The infinite-dimensional disturbance observers as well as the introduction of the continuum cooperative signal observer are contained in Section 6. The networked controller design is

completed with the output feedback regulator design in Section 7. A simulation example demonstrates the results of the article.

Notation: If convenient the (z, t) -dependence is dropped for clarity, that is, $x(0, t) = x(0)$, $x(z, t) = x$, for example.

2 | PROBLEM FORMULATION

Consider the *linear finite-dimensional MAS*

$$\dot{x}_i(t) = u_i(t) + G_i d(t), \quad t > 0, \quad (1)$$

for $i = 0, \dots, N$, $N > 1$, with the state $x_i(t) = \text{col}(x_i^1(t), x_i^2(t)) \in \mathbb{R}^2$ describing the position of agent i in the (x^1, x^2) -plane. The input is $u_i(t) \in \mathbb{R}^2$ and $d(t) \in \mathbb{R}^q$ is an unknown disturbance. The initial condition (IC) of (1) is $x_i(0) = x_{i,0} \in \mathbb{R}^2$.

The disturbances d acting on the agents in (1) are described by the *disturbance model*

$$\dot{v}_d(t) = S_d v_d(t), \quad t > 0, v_d(0) = v_{d,0} \in \mathbb{R}^{n_d}, \quad (2a)$$

$$d(t) = P_d v_d(t), \quad t \geq 0, \quad (2b)$$

in which the matrix $S_d \in \mathbb{R}^{n_d \times n_d}$ has all its eigenvalues on the imaginary axis, that is, the *spectrum* satisfies $\sigma(S_d) \subset j\mathbb{R}$, and $P_d \in \mathbb{R}^{q \times n_d}$ with the pair (P_d, S_d) being observable. Hence, (2) describes a wide class of disturbances including constant, polynomial and trigonometric functions of time as well as linear combinations thereof. The allowed area for containment control in the plane is specified by the positions $r_i(t) \in \mathbb{R}^2$ of the *dynamic leaders*

$$\dot{v}_i(t) = S_i v_i(t), \quad i = 1, \dots, l, l \geq 2, v_i(0) \in \mathbb{R}^{n_i}, \quad (3a)$$

$$r_i(t) = P_i v_i(t), \quad t \geq 0. \quad (3b)$$

Therein, $S_i \in \mathbb{R}^{n_i \times n_i}$, $i = 1, \dots, l$, satisfies $\sigma(S_i) \subset j\mathbb{R}$ and $P_i \in \mathbb{R}^{2 \times n_i}$. In the sequel, the aggregated leader output $r = \text{col}(r_1, \dots, r_l)$ and leader state $v = \text{col}(v_1, \dots, v_l)$ will be used giving rise to

$$\dot{v}(t) = S v(t), \quad t > 0, v(0) = v_0 \in \mathbb{R}^n, \quad (4a)$$

$$r(t) = P v(t), \quad t \geq 0 \quad (4b)$$

with $S = \text{diag}(S_1, \dots, S_l)$, $P = \text{diag}(P_1, \dots, P_l)$ and $n = n_1 + \dots + n_l$.

In this article, the *containment control problem* for the MAS (1) is solved, that is,

$$\lim_{t \rightarrow \infty} \text{dist}(x_i(t), \text{Co}(r_k(t), k = 1, \dots, l)) = 0 \quad (5)$$

for $i = 0, \dots, N$. Therein, $\text{dist}(x, S) = \inf_{y \in S} \|x - y\|$ is the *distance* of a vector $x \in \mathbb{R}^2$ from a set $S \subset \mathbb{R}^2$ in the Euclidean norm and

$$\text{Co}(r_i(t), i = 1, \dots, l) = \left\{ \sum_{i=1}^l c_i r_i(t) \mid r_i(t) \in \mathbb{R}^2, c_i \in \mathbb{R}, c_i \geq 0, \sum_{i=1}^l c_i = 1 \right\} \quad (6)$$

is the *convex hull* spanned by the leaders (3). These properties should hold for all solutions generated by the disturbance model (2) and leaders (3) independent from the ICs of the MAS (1) and of the networked controller. This means that the agents (1) converge to the containment area specified by the convex hull of the leaders in the presence of disturbances. Then, the leaders are able to guide the corresponding agent formation through, for example, a hazardous area.

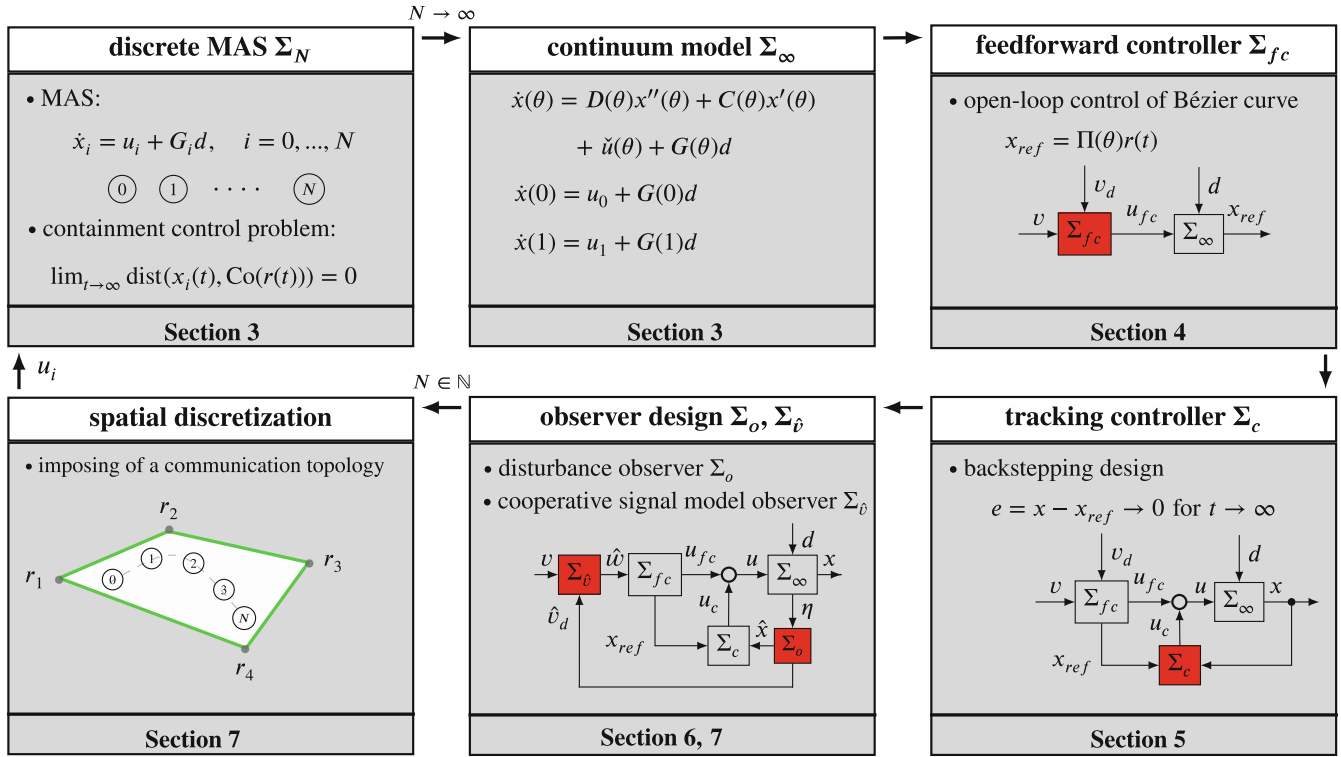


FIGURE 1 Solution of the containment control problem using continuum models and sections, where the corresponding design steps are presented.

In this contribution the containment problem is solved using the networked controller

$$\dot{\hat{w}}_i = S_w \hat{w}_i + \mu \sum_{j=0}^N b_{ij} (\hat{w}_j - \hat{w}_i) \quad (7a)$$

$$u_i^\nu = \sum_{j=0}^N a_{ij}^\nu (x_j^\nu - x_i^\nu) + (m_i^\nu)^\top \hat{w}_i, \quad i = 1, \dots, N-1, \quad \nu = 1, 2 \quad (7b)$$

$$u_i^\nu = \sum_{j=0}^N k_{ij}^\nu (\hat{x}_j^\nu - (\pi_j^\nu)^\top \bar{P} \bar{w}) + (m_i^\nu)^\top \bar{w}_i, \quad i = 0, N, \quad \nu = 1, 2 \quad (7c)$$

with $S_w = \text{diag}(S, S_d)$, $\bar{P} = [P, 0]$, $w(t) = \text{col}(v(t), v_d(t))$, $\bar{w}(t) = \text{col}(v(t), \hat{v}_d(t))$ and $\hat{w}_0(t) = \hat{w}_N(t) = \bar{w}(t)$. Therein $\nu = 1, 2$ describes the x^1 - and x^2 -coordinate. The boundary agents $i = 0, N$, get the estimates for (2) and of the other agent positions (see (7c)) from a disturbance observer designed in Section 6.1. Thereby, the positions of the leaders, for example, the state $v(t)$, are assumed to be available for these agents. Furthermore, it is assumed that the in-domain agents $i = 1, \dots, N-1$ have access to relative state information (see (7b)). Then, the cooperative signal model observer (7a) distributes the leader states $v(t)$ and the estimate of the disturbance state $\hat{v}_d(t)$ from the boundary agents to the in-domain agents. In this article, the networked controller (7) is designed using the continuum formulation, which is independent of the number of agents and for which no communication topology has to be specified a priori. The latter results from a spatial discretization after the design process. Figure 1 shows the design procedure described in the following sections.

3 | OUTPUT REGULATION PROBLEM FOR A CONTINUUM MODEL

The solution of the posed containment control problem is based on a corresponding continuum formulation of the MAS (1) (see Figure 1). To this end, the discrete agent state $x_i(t)$ is rewritten as a continuous agent state $x(\theta, t)$ by exchanging

the agent index i with the continuous variable $\theta \in [0, 1]$. Thus, the discrete MAS (1) turns into its continuum formulation

$$\dot{x}(\theta, t) = u(\theta, t) + G(\theta)d(t), \quad (\theta, t) \in [0, 1] \times \mathbb{R}^+, \quad (8)$$

where $G(\theta)$ is assumed to be piecewise continuous and the initial condition is $x(\theta, 0) = x_0(\theta) \in \mathbb{R}^2$. Then, the continuum formulation of the containment control problem (6) reads

$$\lim_{t \rightarrow \infty} \text{dist}(x(\theta, t), \text{Co}(L_k(t), k = 1, \dots, l)) = 0, \quad \theta \in [0, 1]. \quad (9)$$

To solve the containment control problem (9) cooperatively the diffusive coupling of the agents (see (7b)) has to be introduced in the continuum model (8). To this end, the control law $u(\theta, t) = D(\theta)x_{\theta\theta}(\theta, t) + C(\theta)x_\theta(\theta, t) + \check{u}(\theta, t)$, $\theta \in (0, 1)$, is used, which is the continuum formulation of the right-hand side of (7b). This leads to the *parabolic PDE*

$$\dot{x}(\theta) = D(\theta)x''(\theta) + C(\theta)x'(\theta) + \check{u}(\theta) + G(\theta)d, \quad \theta \in (0, 1) \quad (10a)$$

with $D = \text{diag}(d_1, d_2) \in (C^2[0, 1])^{2 \times 2}$, $\det D(\theta) \neq 0$, $\theta \in [0, 1]$, $C = \text{diag}(c_1, c_2) \in (C^1[0, 1])^{2 \times 2}$, the new input $\check{u}(\theta)$ and the dynamic boundaries

$$\dot{x}(0) = u(0) + G(0)d, \quad (10b)$$

$$\dot{x}(1) = u(1) + G(1)d. \quad (10c)$$

In the next step, it is shown that (9) leads to an output regulation problem for (10), for which a feedforward controller is determined.

4 | FEEDFORWARD CONTROLLER DESIGN

For the feedforward controller design introduce the *tracking error*

$$e(\theta, t) = x(\theta, t) - \Pi(\theta)r(t) \quad (11)$$

with $\Pi(\theta) \in \mathbb{R}^{2 \times 2l}$. By differentiating (11) w.r.t. time and inserting (10) and (3), the *error system*

$$\dot{e}(\theta) = D(\theta)e''(\theta) + C(\theta)e'(\theta), \quad \theta \in (0, 1), \quad (12a)$$

$$\dot{e}(0) = \check{u}(0), \quad (12b)$$

$$\dot{e}(1) = \check{u}(1) \quad (12c)$$

results if the *feedforward controller*

$$\check{u}(\theta) = (\Pi(\theta)PS - D(\theta)\Pi''(\theta)P - C(\theta)\Pi'(\theta)P)v - G(\theta)P_d v_d, \quad \theta \in (0, 1), \quad (13a)$$

$$u(0) = \Pi(0)PSv - G(0)P_d v_d + \check{u}(0), \quad (13b)$$

$$u(1) = \Pi(1)PSv - G(1)P_d v_d + \check{u}(1) \quad (13c)$$

with the new input $\check{u}(\beta, t) \in \mathbb{R}^2$ for $\beta \in \{0, 1\}$ is applied. This controller allows the deployment of any desired agent formation $x_f(\theta, t) = \Pi(\theta)r(t) \in \mathbb{R}^2$ in the plane. In the following it is shown how $\Pi(\theta)$ has to be chosen to ensure that the desired formation lies for all time in the specified containment area. For this, consider the *Bézier curve*

$$\Pi(\theta)r(t) = \sum_{i=0}^n B_i r(t) b_i^n(\theta), \quad \theta \in [0, 1]. \quad (14)$$

The control points $B_i r(t)$ are determined by

$$B_i = \alpha_i^\top \otimes I_2, \quad i = 0, \dots, n \quad (15)$$

and satisfy $\alpha_i = [\alpha_{ij}] \in \mathbb{R}^l$, $\alpha_{ij} \geq 0$, $j = 1, \dots, l$, $\alpha_{i1} + \dots + \alpha_{il} = 1$, in which $\otimes$ denotes the Kronecker product. Furthermore, $b_i^n(\theta)$ in (14) are the Bernstein polynomials

$$b_i^n(\theta) = \binom{n}{i} \theta^i (1 - \theta)^{n-i}, \quad i = 0, \dots, n \quad (16)$$

of degree n . Important properties of Bernstein polynomials for the following are the *partition of unity*

$$1 = \sum_{i=0}^n b_i^n(\theta), \quad \theta \in [0, 1], \quad (17)$$

the *boundary points*

$$b_0^n(0) = 1, \quad b_i^n(0) = 0, \quad i = 1, \dots, n, \quad (18a)$$

$$b_n^n(1) = 1, \quad b_i^n(1) = 0, \quad i = 0, \dots, n-1 \quad (18b)$$

and the *positivity*

$$b_i^n(\theta) \geq 0, \quad \theta \in [0, 1], \quad i = 0, \dots, n. \quad (19)$$

For more details on Bézier curves and their properties the interested reader is referred to Reference 13 (Chapters 5 and 6). The next lemma shows that the agent formation (14) always stays in the containment area.

Lemma 1 (Containment control). *The desired formation profile satisfies $\Pi(\theta)r(t) \in Co(r_i(t), i = 1, \dots, l)$, $t \geq 0$, $\theta \in [0, 1]$.*

Proof. Obviously, the control points satisfy

$$B_i r(t) = (\alpha_i^\top \otimes I_2) r(t) = \sum_{j=1}^l \alpha_{ij} r_j(t) \in C_1 = Co(r_j(t), j = 1, \dots, l), \quad i = 0, \dots, n, t \geq 0, \quad (20)$$

by (15), the properties of α_{ij} and the definition of a convex hull (see Section 2). Furthermore, the Bézier curve itself is always contained in the convex hull spanned by its control points $B_i r(t)$, $i = 0, \dots, n$, (see Figure 2), that is, $\Pi(\theta)r(t) \in C_2 = Co(B_i r(t), i = 0, \dots, n)$, $t \geq 0$, $\theta \in [0, 1]$. This follows from

$$\Pi(\theta)r(t) = \sum_{i=0}^n b_i^n(\theta) B_i r(t) \quad (21)$$

in view of (19) and (17). Consequently, the result of the lemma is implied by $\Pi(\theta)r(t) \in C_2 \subseteq C_1$, $t \geq 0$, $\theta \in [0, 1]$. ■

Note that the control points $B_i r(t)$, $i \in \{0, n\}$, coincide with the position of the boundary agents $x(\theta, t)$, $\theta \in \{0, 1\}$, by (18). Furthermore, the vector $B_1 r(t) - B_0 r(t)$ is the tangent in the starting point $B_0 r(t)$ and accordingly $B_{n-1} r(t) - B_n r(t)$ is the tangent in the endpoint $B_n r(t)$ (cf. Figure 2). Hence, the choice of $B_1 r(t)$ and $B_{n-1} r(t)$ influences the behavior of the formation at the boundaries. The remaining control points $B_i r(t)$, $i = 1, \dots, n-1$, allow to significantly shape the formation at $\theta = i/n$, because the single maximum of the corresponding Bernstein polynomial $b_i^n(\theta)$ is at $\theta = i/n$. This

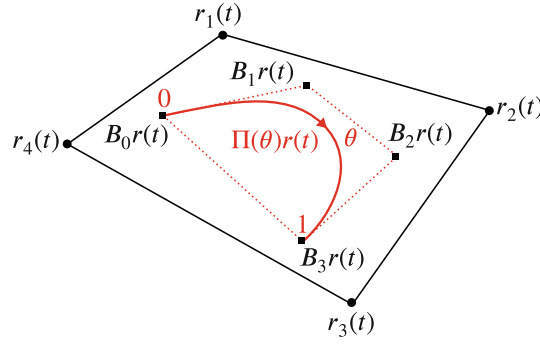


FIGURE 2 Containment area (solid black polygon) specified by the leader positions $r_i(t)$, $i = 1, \dots, 4$. The control points $B_i r(t)$, $i = 0, \dots, 3$, determine the convex hull (dotted red polygon), in which the formation $\Pi(\theta)r(t)$ (red solid line) always lies. The latter is a Bézier curve parameterized by $\theta \in [0, 1]$.

results in a wide class of smooth curves, for which a rich literature exists and corresponding results can be found in Reference 13. Hence, applying the feedforward controller (13) as well as ensuring a stable error system (12) implies

$$\lim_{t \rightarrow \infty} (x(\theta, t) - \Pi(\theta)r(t)) = 0 \quad (22)$$

and thus (9), since $\Pi(\theta)r(t)$ is in the convex hull of the leaders.

To assure asymptotic convergence of the MAS towards the Bézier curve $\Pi(\theta)L(t)$ and therefore (9), the error system (12) has to be stable. To this end, $D(\theta)$ and $C(\theta)$ can be chosen accordingly and the boundary dynamics (12b), (12c) have to be stabilized, as shown in Reference 7. This leads to a restriction of the resulting communication topology. Therefore, in the following section a boundary feedback controller is developed, which for an arbitrary choice of $D(\theta)$ and $C(\theta)$, leading possibly to an unstable PDE (12a) solves the posed containment control problem. Thus, the specification of the communication topology is completely decoupled from the stabilization and the convergence rate can be simply assigned. Furthermore, the control effort is specifically distributable between the boundary agents. Also, the presented boundary stabilization allows the scalable design of a continuum disturbance observer requiring the knowledge of only one agent position. To this end, in the next section a tracking controller is determined to ensure (22) using backstepping (see Figure 1).

5 | STATE FEEDBACK TRACKING CONTROLLER

In order to simplify the subsequent design, the *Hopf-Cole-type transformation*

$$\bar{e} = T(\theta)e = \begin{bmatrix} T_1(\theta) \\ T_2(\theta) \end{bmatrix} e \quad (23)$$

with $T_i(\theta) = \exp(\int_0^\theta (c_i(\bar{\theta})/(2d_i(\bar{\theta})))d\bar{\theta}) \in \mathbb{R}$, $i = 1, 2$, is applied to eliminate the convection term yielding

$$\dot{\bar{e}} = D(\theta)\bar{e}'' + H(\theta)\bar{e}, \quad (24a)$$

$$\dot{\bar{e}}(0) = \bar{u}(0), \quad (24b)$$

$$\dot{\bar{e}}(1) = \bar{u}(1) \quad (24c)$$

with $H(\theta) = \text{diag}(h_1(\theta), h_2(\theta))$, $h_i(\theta) = -d_i(\theta)(c_i^2(\theta)/(4d_i^2(\theta)) + (c_i(\theta)/(2d_i(\theta)))')$, $i = 1, 2$, and $\bar{u}(\beta, t) = T(\beta)\bar{u}(\beta, t) \in \mathbb{R}^2$, $\beta \in \{0, 1\}$. In order to fully exploit the design freedom for stabilizing the error system (12), a bilateral backstepping controller is determined (for related results see References 14, 15, and 17). This problem can be traced back to an unilateral design by introducing the invertible *folding transformation*

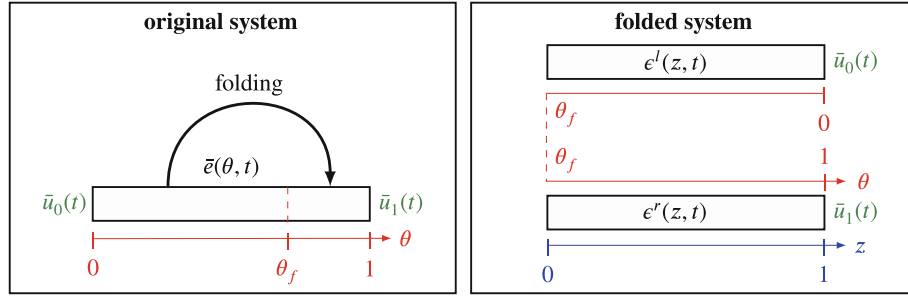


FIGURE 3 The original system is transformed into the folded system using the folding transformation (25). Therefore, the spatial domain θ is kinked at the folding point θ_f resulting in an unilateral actuation at $z = 1$.

$$z = f(\theta) = \begin{cases} (\theta_f - \theta)/\theta_f, & \theta < \theta_f \\ (\theta - \theta_f)/(1 - \theta_f), & \theta \geq \theta_f \end{cases}, \quad (25)$$

where $\theta = \theta_f \in (0, 1)$ is the *folding point* at which the spatial coordinate $\theta \in [0, 1]$ is folded into the domain $z \in [0, 1]$ describing the corresponding spatial domain left and right of the folding point (see Figure 3). The latter can be utilized to distribute the control effort between the inputs $\bar{u}(\beta)$, $\beta \in \{0, 1\}$. With this, the new state

$$\varepsilon(z, t) = \begin{bmatrix} \varepsilon_1(z, t) \\ \varepsilon_2(z, t) \end{bmatrix} = \begin{bmatrix} \bar{e}(\theta_f - \theta_f z, t) \\ \bar{e}(\theta_f + (1 - \theta_f)z, t) \end{bmatrix} \in \mathbb{R}^4, \quad (26)$$

where $\varepsilon_i \in \mathbb{R}^2$, $i = 1, 2$, is introduced, that describes the error system (12) at the left and right hand side of the folding point. Then, the resulting folded system leads to a 4×4 parabolic system governed by the PDE

$$\dot{\varepsilon} = \Lambda(z)\varepsilon'' + A(z)\varepsilon, \quad z \in (0, 1) \quad (27a)$$

with $\Lambda(z) = \text{diag}(\lambda_1(z), \dots, \lambda_4(z))$ and $\lambda_i(z) = d_i(\theta_f - \theta_f z)/\theta_f^2$, $i = 1, 2$, $\lambda_i(z) = d_{i-2}(\theta_f + (1 - \theta_f)z)/(1 - \theta_f)^2$, $i = 3, 4$. Thereby, $\lambda_1(z) > \dots > \lambda_4(z)$ is assumed, which can always be satisfied by a suitable choice of the folding point θ_f . In the case of a constant Λ the folding point can be chosen arbitrarily (see Reference 15). Furthermore, $A(z) = \text{diag}(a_1(z), \dots, a_4(z))$ with $a_i(z) = h_i(\theta_f - \theta_f z)$, $i = 1, 2$, and $a_i(z) = h_{i-2}(\theta_f + (1 - \theta_f)z)$, $i = 3, 4$. At the boundaries of (27a) the ODE dynamics

$$\dot{\varepsilon}(1) = v \quad (27b)$$

with $v(t) = \text{col}(\bar{u}(0, t), \bar{u}(1, t)) \in \mathbb{R}^4$ are attached. The PDEs in (27a) are coupled at the folding point $z = 0$ via the *folding BCs*

$$\varepsilon_1'(0) = -\tilde{\theta}_f \varepsilon_2'(0), \quad (27c)$$

$$\varepsilon_2(0) = \varepsilon_1(0) \quad (27d)$$

and $\tilde{\theta}_f = \theta_f/(1 - \theta_f)^2$, which have to be introduced for ensuring the regularity of the solution.

For mapping (27) into a stable target system, consider the transformations

$$\tilde{\varepsilon} = \begin{bmatrix} \tilde{\varepsilon}_1 \\ \tilde{\varepsilon}_2 \end{bmatrix} = \varepsilon - \int_0^z K(z, \zeta) \varepsilon(\zeta) d\zeta, \quad (28a)$$

$$\bar{\varepsilon}_2 = \tilde{\varepsilon}_2 - \underbrace{\int_0^z Q(z, \zeta) \tilde{\varepsilon}_2(\zeta) d\zeta}_{\tau_c[\tilde{\varepsilon}_2]} - \int_0^1 P(z, \zeta) \tilde{\varepsilon}_1(\zeta) d\zeta - \Gamma(z) \tilde{\varepsilon}_1(1) \quad (28b)$$

with the kernels $K(z, \zeta) \in \mathbb{R}^{4 \times 4}$, $P(z, \zeta)$, $Q(z, \zeta) \in \mathbb{R}^{2 \times 2}$ and $\Gamma(z) \in \mathbb{R}^{2 \times 2}$. The invertible backstepping transformation (28a) maps (27) into the *intermediate target system*

$$\dot{\tilde{\varepsilon}} = \Lambda(z)\tilde{\varepsilon}'' - \mu_c \tilde{\varepsilon} - \tilde{A}_0(z)\tilde{\varepsilon}(0) - \tilde{A}_1(z)\tilde{\varepsilon}'(0), \quad (29a)$$

$$\tilde{\varepsilon}'_1(0) = -\tilde{\theta}_f \tilde{\varepsilon}'_2(0), \quad \tilde{\varepsilon}_2(0) = \tilde{\varepsilon}_1(0), \quad (29b)$$

$$\dot{\tilde{\varepsilon}}_1(1) = \mu_0 \tilde{\varepsilon}_1(1), \quad \dot{\tilde{\varepsilon}}_2(1) = \tilde{v} \quad (29c)$$

with $\mu_c \in \mathbb{R}$, strictly lower triangular matrices

$$\tilde{A}_i(z) = \begin{bmatrix} \tilde{A}_i^1(z) & 0 \\ \tilde{A}_i^2(z) & 0 \end{bmatrix} \in \mathbb{R}^{4 \times 4}, \quad i = 0, 1 \quad (30)$$

(see also Reference 15) and $\mu_0 < 0$. Therein, the new input $\tilde{v}(t) \in \mathbb{R}^2$ is defined by

$$\begin{aligned} v &= \int_0^1 K(1, \zeta) \Lambda(\zeta) \varepsilon''(\zeta) d\zeta + \int_0^1 K(1, \zeta) A(\zeta) \varepsilon(\zeta) d\zeta + \begin{bmatrix} \mu_0 \varepsilon_1(1) - \mu_0 \int_0^1 E_1^\top K(1, \zeta) \varepsilon(\zeta) d\zeta \\ \tilde{v} \end{bmatrix} \\ &= \mathcal{K}_1[\varepsilon, \varepsilon'(0), \varepsilon'(1), \varepsilon(0), \varepsilon(1), \tilde{v}]. \end{aligned} \quad (31)$$

The latter result follows from making use of (27) and two integration by parts. In order to determine the backstepping transformation (28a), it can be shown using the results in Reference 15 that the kernel $K(z, \zeta)$ has to satisfy the *kernel equations*

$$\Lambda(z)K_{zz}(z, \zeta) - (K(z)\Lambda)_{\zeta\zeta}(\zeta) = K(z, \zeta)(A(\zeta) + \mu_c I), \quad (z, \zeta) \in \{0 < \zeta < z < 1\}, \quad (32a)$$

$$\Lambda(z)K'(z, z) + \Lambda(z)K_z(z, z) + (K(z)\Lambda)_\zeta(z) = -(A(z) + \mu_c I), \quad (32b)$$

$$K(z, z)\Lambda(z) - \Lambda(z)K(z, z) = 0, \quad K(0, 0) = 0, \quad (32c)$$

$$K(z, 0)\Lambda(0)((1/\tilde{\theta}_f)E_2 - E_1) = -\tilde{A}_1(z), \quad (32d)$$

$$(K(z)\Lambda)_\zeta(0)(E_1 + E_2) = \tilde{A}_0(z) \quad (32e)$$

with $E_1 = [I_2, 0]^\top$ and $E_2 = [0, I_2]^\top$. Different from the usual kernel equations found for coupled parabolic PDEs, the kernel equations (32) are subject to the folding BCs (32d) and (32e). This significantly impedes their solution. In References 14 and 15 it is verified that (32) have a piecewise C^2 -solution after introducing artificial BCs and also a constructive solution procedure is provided.

Since the $\tilde{\varepsilon}_1$ - and $\tilde{\varepsilon}_2$ -systems in (29) are still bidirectionally coupled, the decoupling transformation (28b) is applied. With $\bar{\varepsilon}_1 = \tilde{\varepsilon}_1$ this leads to the *final target system*

$$\dot{\bar{\varepsilon}} = \Lambda(z)\bar{\varepsilon}'' - \mu_c \bar{\varepsilon} - \bar{A}_0(z)\bar{\varepsilon}(0) - \bar{A}_1(z)\bar{\varepsilon}'(0), \quad (33a)$$

$$\bar{\varepsilon}'_1(0) = -\tilde{\theta}_f \bar{\varepsilon}'_2(0), \quad \bar{\varepsilon}_2(0) = \bar{\varepsilon}_1(0), \quad (33b)$$

$$\dot{\bar{\varepsilon}}_1(1) = \mu_0 \bar{\varepsilon}_1(1), \quad \dot{\bar{\varepsilon}}_2(1) = \mu_1 \bar{\varepsilon}_2(1), \quad (33c)$$

where $\mu_i < 0$, $i = 0, 1$, and blockdiagonal matrices

$$\bar{A}_i(z) = \begin{bmatrix} \bar{A}_i^1(z) & 0 \\ 0 & \bar{A}_i^2(z) \end{bmatrix} \in \mathbb{R}^{4 \times 4}, \quad i = 0, 1. \quad (34)$$

Thereby, the state feedback

$$\begin{aligned}\tilde{v} &= \int_0^1 P(1, \zeta) \dot{\tilde{\varepsilon}}_1(\zeta) d\zeta + \int_0^1 (Q(1, \zeta) \dot{\tilde{\varepsilon}}_2(\zeta) d\zeta - \mu_1(-\tilde{\varepsilon}_2(1)) \\ &\quad + \int_0^1 (P(1, \zeta) \tilde{\varepsilon}_1(\zeta, t) + Q(1, \zeta) \tilde{\varepsilon}_2(\zeta, t)) d\zeta + \Gamma(1) \tilde{\varepsilon}_1(1)) + \mu_0 \Gamma(1) \tilde{\varepsilon}_1(1) \\ &= \mathcal{K}_2[\tilde{\varepsilon}, \tilde{\varepsilon}(1), \tilde{\varepsilon}'_1(0), \tilde{\varepsilon}'_2(1), \tilde{\varepsilon}_1(0), \tilde{\varepsilon}_2(1)]\end{aligned}\quad (35)$$

is employed, which follows from (33) and two integrations by parts. Using the results in Reference 15 and introducing $\Lambda = \text{diag}(\Lambda_1, \Lambda_2)$, $\Lambda_i \in \mathbb{R}^{2 \times 2}$, $i = 1, 2$, it is straightforward to show that the kernels $P(z, \zeta)$ and $Q(z, \zeta)$ appearing in the decoupling transformation (28b) have to solve the *kernel equations*

$$\Lambda_2(z) P_{zz}(z, \zeta) - (P(z) \Lambda_1)_{\zeta\zeta}(\zeta) = 0, \quad (z, \zeta) \in (0, 1)^2, \quad (36a)$$

$$P(0, \zeta) = P_z(0, \zeta) = P(z, 1) = 0, \quad (36b)$$

$$\Lambda_2(z) Q_{zz}(z, \zeta) - (Q(z) \Lambda_2)_{\zeta\zeta}(\zeta) = 0, \quad (z, \zeta) \in \{0 < \zeta < z < 1\}, \quad (36c)$$

$$\Lambda_2(z) Q'(z, z) + \Lambda_2(z) Q_z(z, z) + (Q(z) \Lambda_2)_\zeta(z) = 0, \quad (36d)$$

$$Q(z, z) \Lambda_2(z) - \Lambda_2(z) Q(z, z) = 0, \quad Q(0, 0) = 0, \quad (36e)$$

$$P(z, 0) \Lambda_1(0) - \frac{1}{\theta_f} Q(z, 0) \Lambda_2(0) = \mathcal{T}_c[\tilde{A}_1^2](z) - \int_0^1 P(z, \zeta) \tilde{A}_1^1(\zeta) d\zeta + \frac{1}{\theta_f} \tilde{A}_1^2(z), \quad (36f)$$

$$(Q(z) \Lambda_2)_\zeta(0) + (P(z) \Lambda_1)_\zeta(0) = -\mathcal{T}_c[\tilde{A}_0^2](z) + \tilde{A}_0^2(z) + \int_0^1 P(z, \zeta) \tilde{A}_0^1(\zeta) d\zeta. \quad (36g)$$

Since (36) consists of bidirectionally coupled PDEs, they have to be solved simultaneously. For this, it is shown in Reference 15 that the method of successive approximations can be utilized to verify that a piecewise continuous solution $P(z, \zeta)$ and $Q(z, \zeta)$ exists. Furthermore, $\Gamma(z)$ is the solution of the *decoupling equations*

$$\Lambda_2(z) \Gamma''(z) - (\mu_0 + \mu_c) \Gamma(z) = -(P(z) \Lambda_1)_\zeta(1), \quad z \in (0, 1], \quad (37a)$$

$$\Gamma(0) = \Gamma'(0) = 0, \quad (37b)$$

where a C^2 -solution $\Gamma(z)$ is readily attainable.

Unfolding (33) with

$$\tilde{\varepsilon}(\theta, t) = \begin{cases} \tilde{\varepsilon}_1((\theta_f - \theta)/\theta_f, t), & \theta < \theta_f \\ \tilde{\varepsilon}_2((\theta_f - \theta_f)/(1 - \theta_f), t), & \theta \geq \theta_f \end{cases} \quad (38)$$

and

$$\check{A}_0(\theta) = \begin{cases} \tilde{A}_0^1((\theta_f - \theta)/\theta_f), & \theta < \theta_f \\ \tilde{A}_0^2((\theta_f - \theta_f)/(1 - \theta_f)), & \theta \geq \theta_f \end{cases}, \quad \check{A}_1(\theta) = \begin{cases} -\theta_f \tilde{A}_1^1((\theta_f - \theta)/\theta_f), & \theta < \theta_f \\ (1 - \theta_f) \tilde{A}_1^2((\theta_f - \theta_f)/(1 - \theta_f)), & \theta \geq \theta_f \end{cases} \quad (39)$$

directly yields the *unfolded target system*

$$\dot{\tilde{e}} = D(\theta) \tilde{e}'' - \mu_c \tilde{e} - \check{A}_0(\theta) \tilde{e}(\theta_f) - \check{A}_1(\theta) \tilde{e}'(\theta_f), \quad (\theta, t) \in (0, 1) \times \mathbb{R}^+, \quad (40a)$$

$$\dot{\tilde{e}}(0) = \mu_0 \tilde{e}(0), \quad (40b)$$

$$\dot{\tilde{e}}(1) = \mu_1 \tilde{e}(1) \quad (40c)$$

related to (24). Therein, $\tilde{A}_i(\theta) \in \mathbb{R}^{4 \times 4}$, $i = 0, 1$, are strictly lower triangular matrices so that (40) is a cascade of parabolic PDE-ODE systems. The next theorem verifies closed-loop stability of the tracking error dynamics.

Theorem 1 (Closed-loop stability). *Suppose that $\lambda_{\max}$ is the maximum eigenvalue of the operator $\mathcal{A}h = Dh''$, $h \in D(\mathcal{A}) = \{h \in (H^2(0, 1))^2 \mid h(0) = h(1) = 0\}$. If $\mu_c > \lambda_{\max}$ and $\mu_i < 0$, $i = 0, 1$, then the error system (12) with the state $e_c(t) = \text{col}(e(0, t), e(1, t), e(t))$, $e(t) = e(\theta, t)$, $\theta \in [0, 1]$, and (31), (35) is exponentially stable in the norm*

$$\|\cdot\|_c = \left(\|\cdot\|_{\mathbb{C}^2}^2 + \|\cdot\|_{\mathbb{C}^2}^2 + \|\cdot\|_{L_2^2}^2 \right)^{\frac{1}{2}} \quad (41)$$

with the decay rate $\alpha_c = -\max(\mu_0, \mu_1, \lambda_{\max} - \mu_c) > 0$, that is,

$$\|e_c(t)\|_c \leq M_c e^{(-\alpha_c + c)t} \|e_c(0)\| \quad t \geq 0 \quad (42)$$

for all $e_c(0) \in \mathbb{C}^2 \oplus \mathbb{C}^2 \oplus (H^2(0, 1))^2 \subset X$ compatible with the boundary conditions of the controlled plant, $M_c \geq 1$ and any $c > 0$ holds.

Proof. The final target system can be written as the ODE-PDE-ODE-system

$$\dot{\varphi}_0 = \mu_0 \varphi_0, \quad (43a)$$

$$\dot{\varphi}_1 = \mu_1 \varphi_1, \quad (43b)$$

$$\dot{\tilde{e}} = D(\theta)\tilde{e}'' - \mu_c \tilde{e} - \bar{A}_0(\theta)\tilde{e} - \bar{A}_1(\theta)\tilde{e}', \quad (43c)$$

$$\tilde{e}(0) = \varphi_0, \quad (43d)$$

$$\tilde{e}(1) = \varphi_1 \quad (43e)$$

with the ODE-states $\varphi_0(t) \in \mathbb{R}^2$ and $\varphi_1(t) \in \mathbb{R}^2$. In this representation the results from Reference 18 can directly be applied, resulting in the decay rate $\alpha_c = -\max(\mu_0, \mu_1, \lambda_{\max} - \mu_c) > 0$. The exponential stability of (12) with the control laws (31) and (35) applied, follows due to the bounded invertibility of the transformations (28). In the next section a disturbance observer and a cooperative signal model observer is designed (see Figure 1). ■

6 | OBSERVER DESIGN

After a spatial discretization and introducing the joint signal model

$$\dot{w} = S_w w, \quad t > 0, w(0) = w_0 \in \mathbb{R}^{n_w} \quad (44)$$

with $w = \text{col}(v, v_d)$, $S_w = \text{diag}(S, S_d) \in \mathbb{R}^{n_w \times n_w}$, $n_w = n + n_d$, the networked controller designed so far takes the form

$$u_i^v = \sum_{j=0}^N a_{ij}^v (x_j^v - x_i^v) + (m_i^v)^\top w, \quad i = 1, \dots, N-1, \quad (45a)$$

$$u_i^v = \sum_{j=0}^N k_{ij}^v (x_j^v - (\pi_j^v)^\top r) + (m_i^v)^\top w, \quad i = 0, N \quad (45b)$$

with $v = 1, 2$. Obviously, an implementation of (45b) requires the knowledge of the agent positions $x_i(t)$, $i = 0, \dots, N$ and the signal model states $w(t)$. In contrast, only relative state information is required in (45a), which is supposed to be available for the in-domain agents. In particular, this state feedback results from the introduction of the diffusive and advective coupling in Section 3. In the sequel, it is assumed that the boundary agents have direct access to the leader

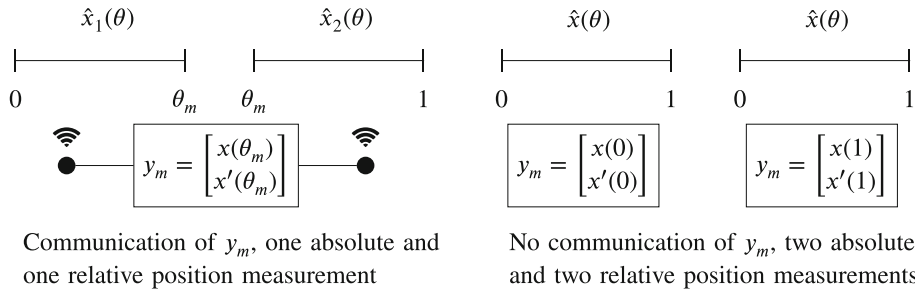


FIGURE 4 On the left hand side an observer using the folding approach is shown. In this case only one absolute and one relative measurement is needed. Both boundary agents reconstruct the state information $\hat{x}_1(\theta)$, $\hat{x}_2(\theta)$ between them and the measured agent at $\theta_m \in (0, 1)$. On the right hand side two absolute and two relative position measurements are required to reconstruct the full state $\hat{x}(\theta)$ for both observers.

positions $r(t)$ so that the state $v(t)$ is known to the boundary agents. Then, a continuum disturbance observer is designed to estimate the agent positions and the state $v_d(t)$ of the disturbance model. For this, the two approaches shown in Figure 4 are possible. The first possibility is to design two observers similar to References 19 (Chapter 4.11) and 14 using a folding approach shown on the left hand side of Figure 4. Here only one absolute and one relative position measurement of an arbitrary in-domain agent is required. Using this information both boundary agents reconstruct only the states between them and the measured agent in the intervals $[0, \theta_m]$ and $[\theta_m, 1]$. This observer, however, requires the communication of the measurement to the boundary agents. It should be noted that the choice $\theta_m = \theta_f$ is advantageous, because then only the in-domain measurement has to be communicated and no further state estimates. This can directly be inferred from the folding approach in Section 5. The other possibility is to determine one disturbance observer for each boundary agent estimating the whole state $\hat{x}(\theta)$ (right hand side of Figure 4). Then, each boundary agent has to measure its absolute position and its position relative to its neighbor. In this case no communication is required.

As a result an implementable tracking controller is attained, since only the available measurement is used in the feedback (see Figure 1).

6.1 | Disturbance observer

6.1.1 | Folding observer

For the design of the folding observer the *in-domain measurement*

$$\eta = \begin{bmatrix} x(\theta_m) \\ x'(\theta_m) \end{bmatrix}, \quad \theta_m \in (0, 1) \quad (46)$$

is considered. In Reference 14 it is shown that the corresponding observer design can be traced back to unilateral backstepping by making use of a folding at $\theta = \theta_m$. With the disturbance model (2), applying the Hopf-Cole-type transformation $\bar{x} = T(\theta)x$ and $T(\theta) = \text{diag}(T_1(\theta), T_2(\theta))$ (cf. (23)) and defining the folded states

$$\xi(z, t) = \begin{bmatrix} \xi_1(z, t) \\ \xi_2(z, t) \end{bmatrix} = \begin{bmatrix} \bar{x}(\theta_m - \theta_m z, t) \\ \bar{x}(\theta_m + (1 - \theta_m)z, t) \end{bmatrix} \in \mathbb{R}^4, \quad z \in [0, 1] \quad (47)$$

(cf. (26)) it is straightforward to verify that the folded continuum model (10) can be represented by

$$\dot{v}_d = S_d v_d, \quad (48a)$$

$$\dot{\xi} = \Lambda(z)\xi'' + A(z)\xi + \check{u}(z) + \check{G}(z)P_d v_d, \quad (z, t) \in (0, 1) \times \mathbb{R}^+, \quad (48b)$$

$$\xi'_1(0) = -\tilde{\theta}_m \xi'_2(0), \quad \xi_2(0) = \xi_1(0), \quad (48c)$$

$$\dot{\xi}(1) = u_1 + G_1 P_d v_d. \quad (48d)$$

Therein, $\tilde{\theta}_m = \theta_m / (1 - \theta_m)^2$ and the folded inputs are denoted by $\check{u}$ and u_1 for convenience. They directly result from (13) and applying the folding transformation (25). In (48b) the matrices $\Lambda(z)$ and $A(z)$ are defined in Section 5, while with $G(z) = \text{col}(g_1^\top(z), \dots, g_4^\top(z))$ the result

$$\check{G}(z) = \text{col}(\check{g}_1^\top(z), \dots, \check{g}_4^\top(z)) = \begin{cases} T_1(\theta_f - \theta_f z) g_i^\top(\theta_f - \theta_f z), & i = 1, 2, \\ T_2(\theta_f + (1 - \theta_f z)) g_{i-2}^\top(\theta_f + (1 - \theta_f z)), & i = 3, 4, \end{cases} \quad (49a)$$

$$G_1 = \begin{bmatrix} T(0)G(0) \\ T(1)G(1) \end{bmatrix} \quad (49b)$$

follows and the folded measurement reads

$$\eta = \begin{bmatrix} \xi(0) \\ \xi'(0) \end{bmatrix}. \quad (50)$$

Since $\Lambda = \text{diag}(\Lambda_1, \Lambda_2)$ and $A = \text{diag}(A_1, A_2)$, $A_i(z) \in \mathbb{R}^{2 \times 2}$, $i = 1, 2$, are diagonal matrices, the observer design for (48) can be split into two equal separate *disturbance observers*

$$\dot{\hat{v}}_{d,i} = S_d \hat{v}_{d,i} + L_{d,i}(\xi'_i(0) - \hat{\xi}'_i(0)), \quad (51a)$$

$$\dot{\hat{\xi}}_i = \Lambda_i(z) \hat{\xi}_i'' + A_i(z) \hat{\xi}_i + \check{u}^i(z) + \bar{G}^i(z) \hat{v}_{d,i} + L_i(z)(\xi'_i(0) - \hat{\xi}'_i(0)), \quad (z, t) \in (0, 1) \times \mathbb{R}^+, \quad (51b)$$

$$\hat{\xi}_i(0) = \eta_i, \quad (51c)$$

$$\hat{\xi}_i(1) = u_1^i + \bar{G}_1^i \hat{v}_{d,i} + L_{1,i}(\xi'_i(0) - \hat{\xi}'_i(0)) \quad (51d)$$

with the *observer gains* $L_{d,i} \in \mathbb{R}^{n_d \times 2}$, $L_i(z) \in \mathbb{R}^{2 \times 2}$ and $L_{1,i} \in \mathbb{R}^{2 \times 2}$ as well as the ICs $\hat{v}_{d,i}(0) \in \mathbb{R}^{n_d}$, $\hat{\xi}_i(z, 0) \in \mathbb{R}^2$, $z \in [0, 1]$.

Thereby, in (51) the partitions $\check{u}^i = E_i^\top \check{u}$, $u_1^i = E_i^\top u_1$, $\bar{G}^i = E_i^\top \bar{G} P_d$, and $\bar{G}_1^i = E_i^\top G_1 P_d$, $i = 1, 2$, are utilized. Hence, for each of the controlled agents at the boundaries of the continuum model, that is,

$$\dot{x}_i = u_i + G_i d, \quad i = 0, N, \quad (52)$$

in (1), a disturbance observer (51) is designed (see Figure 4 at the left hand side). In particular, the observer (51) for $i = 0$ is used for agent 0 and the observer for $i = 1$ for agent N .

For determining the observer gains in (51), the index i is dropped to increase readability. Then, introducing the observer errors $e_d = v_d - \hat{v}_d$, $e_\xi = \xi - \hat{\xi}$ yields the *observer error dynamics*

$$\dot{e}_d = S_d e_d - L_d e'_\xi(0), \quad (53a)$$

$$\dot{e}_\xi = \Lambda(z) e''_\xi + A(z) e_\xi + \bar{G}(z) e_d - L(z) e'_\xi(0), \quad (53b)$$

$$e_\xi(0) = 0, \quad (53c)$$

$$\dot{e}_\xi(1) = \bar{G}_1 e_d - L_1 e'_\xi(0). \quad (53d)$$

In order to map (53) into a stable target dynamics, consider the transformations

$$\bar{e}_\xi(1) = e_\xi(1) - Me_d - \int_0^1 N(\zeta)e_\xi(\zeta)d\zeta, \quad (54a)$$

$$e_\xi = \tilde{e}_\xi - \int_0^z \bar{K}(z, \zeta)\tilde{e}_\xi(\zeta)d\zeta = \mathcal{T}_0^{-1}[\tilde{e}], \quad (54b)$$

$$\check{e}_\xi = \tilde{e}_\xi - R(z)e_d \quad (54c)$$

with $M, R(z) \in \mathbb{R}^{2 \times n_d}$ and $N(z), \bar{K}(z, \zeta) \in \mathbb{R}^{2 \times 2}$. The transformation (54a) decouples the $e(1)$ -system in (53d) to obtain the *first intermediate target system*

$$\dot{\bar{e}}_\xi(1) = \bar{\mu}_1 \bar{e}_\xi(1), \quad (55a)$$

$$\dot{e}_d = S_d e_d - L_d e'_\xi(0), \quad (55b)$$

$$\dot{e}_\xi = \Lambda(z)e''_\xi + A(z)e_\xi + \bar{G}(z)e_d - L(z)e'_\xi(0), \quad (55c)$$

$$e_\xi(0) = 0, \quad (55d)$$

$$e_\xi(1) = \bar{e}_\xi(1) + Me_d + \int_0^1 N(\zeta)e_\xi(\zeta)d\zeta \quad (55e)$$

with $\bar{\mu}_1 < 0$, which stabilizes the $\bar{e}_\xi(1)$ -system. Differentiating (54a) as well as inserting (53) and (55) yields after a direct calculation the *decoupling equations*

$$(N(z)\Lambda(z))'' - \bar{\mu}_1 N(z) + N(z)A(z) = 0, \quad z \in [0, 1), \quad (56a)$$

$$N(1)\Lambda(1) = 0, \quad (56b)$$

$$(N\Lambda)'(1) = \bar{\mu}_1 I \quad (56c)$$

and

$$M = \left(\bar{G}_1 - \int_0^1 N(\zeta)\bar{G}(\zeta)d\zeta \right) (S_d - \bar{\mu}_1 I)^{-1}, \quad (57a)$$

$$L_1 = ML_d + \int_0^1 N(\zeta)L(\zeta)d\zeta + N(0)\Lambda(0). \quad (57b)$$

Note that therein the inverse matrix exists, because of $\bar{\mu}_1 < 0$ and $\sigma(S_d) \subset j\mathbb{R}$. It is readily verified that the IVP (56a)–(56c) has a unique C^2 -solution.

In the next step, the invertible backstepping transformation (54b) is applied to map (55) into the *second intermediate target system*

$$\dot{\tilde{e}}_\xi(1) = \bar{\mu}_1 \tilde{e}_\xi(1), \quad (58a)$$

$$\dot{e}_d = S_d e_d - L_d \tilde{e}'_\xi(0), \quad (58b)$$

$$\dot{\tilde{e}}_\xi = \Lambda(z)\tilde{e}''_\xi - \mu_0 \tilde{e}_\xi + \tilde{G}(z)e_d - \tilde{L}(z)\tilde{e}'_\xi(0), \quad (58c)$$

$$\tilde{e}_\xi(0) = 0, \quad (58d)$$

$$\tilde{e}_\xi(1) = \bar{e}_\xi(1) + Me_d \quad (58e)$$

with $\mu_0 \in \mathbb{R}$, the transformed disturbance input $\tilde{G}(z) = \mathcal{T}_0[\bar{G}](z)$ and the auxiliary observer gain $\tilde{L}(z) \in \mathbb{R}^{2 \times 2}$ defined by

$$L(z) = \mathcal{T}_0^{-1}[\tilde{L}](z) - \bar{K}(z, 0)\Lambda(0). \quad (59)$$

Differentiating (54b) w.r.t. time, using (55) and (58) yields the *kernel equations*

$$\Lambda(z)\bar{K}_{zz}(z, \zeta) - (\bar{K}(z)\Lambda)_{\zeta\zeta}(\zeta) = (A(\zeta) - \mu_0 I)\bar{K}(z, \zeta), \quad (z, \zeta) \in \{0 < \zeta < z < 1\}, \quad (60a)$$

$$\Lambda(z)\bar{K}'(z, z) + \Lambda(z)\bar{K}_z(z, z) + (\bar{K}(z)\Lambda)_{\zeta}(z) = \mu_0 I + A(z), \quad (60b)$$

$$\bar{K}(1, \zeta) = -N(\zeta) + \int_{\zeta}^1 N(\bar{\zeta})\bar{K}(\bar{\zeta}, \zeta)d\bar{\zeta}, \quad (60c)$$

$$\Lambda(z)\bar{K}(z, z) - \bar{K}(z, z)\Lambda(z) = 0, \quad (60d)$$

where two integrations by parts and an interchange of the order of integration were applied. The kernel equations (60) coincide with the kernel equations investigated in References 20 and 21 that result after a simple duality transformation (see Reference 22). Thereby, the additional Volterra boundary integral term poses no problem for the well-posedness proof. Hence, the kernel equations (60) admit a unique C^2 -solution, after introducing suitable artificial BCs (see References 20 and 21).

In the last step, the *decoupling transformation* (61e) maps (58) into the *final target system*

$$\dot{\bar{e}}_{\xi}(1) = \bar{\mu}_1 \bar{e}_{\xi}(1), \quad (61a)$$

$$\dot{\check{e}}_{\xi} = \Lambda(z)\check{e}_{\xi}'' - \mu_0 \check{e}_{\xi}, \quad (61b)$$

$$\check{e}_{\xi}(0) = 0, \quad (61c)$$

$$\check{e}_{\xi}(1) = \bar{e}_{\xi}(1), \quad (61d)$$

$$\dot{e}_d = (S_d - L_d R'(0))e_d - L_d \check{e}_{\xi}'(0), \quad (61e)$$

which is a parabolic ODE-PDE-ODE cascade. For this,

$$\tilde{L}(z) = R(z)L_d \quad (62)$$

has to hold and $R(z)$ has to satisfy the *decoupling equations*

$$R(z)S_d - \Lambda(z)R''(z) + \mu_0 R(z) = \bar{G}(z), \quad z \in (0, 1), \quad (63a)$$

$$R(0) = 0, \quad (63b)$$

$$R(1) = M. \quad (63c)$$

The next lemma states the condition for the stabilization of the e_d -system (61e).

Lemma 2 (Observability). *The decoupling equations (63) have a unique C^2 -solution if any eigenvalue of S_d is not an eigenvalue of the PDE subsystem (61b)–(61d). Let $\Psi(z, \zeta, s) : [0, 1]^2 \times \mathbb{C} \rightarrow \mathbb{C}^{4 \times 4}$ be the fundamental matrix solving the IVP*

$$\partial_{\zeta}\Psi(z, \zeta, s) = \begin{bmatrix} 0 & I \\ (s + \mu_0)\Lambda^{-1}(z) & 0 \end{bmatrix} \Psi(z, \zeta, s), \quad \Psi(\zeta, \zeta, s) = I. \quad (64)$$

Then, the pair $(R'(0), S_d)$ is observable iff

$$\left(M - \int_0^1 E_1^\top \Psi(1, \zeta, \nu_i) E_2 \Lambda^{-1}(\zeta) \bar{G}(\zeta) d\zeta \right) \varphi_{i(1)} \neq 0, \quad i = 1, \dots, \rho_d, \quad (65)$$

where $\varphi_{i(1)}$ are the linearly independent eigenvectors w.r.t. the ρ_d distinct eigenvalues ν_i of S_d .

Proof. Assume that the matrix $S_d \in \mathbb{R}^{n_d \times n_d}$ has the Jordan blocks $J_i, i = 1, \dots, \rho_d$, to which the Jordan chains

$$S_d \varphi_{i(1)} = \nu_i \varphi_{i(1)}, \quad (66a)$$

$$S_d \varphi_{i(k)} = \nu_i \varphi_{i(k)} + \varphi_{i(k-1)}, \quad k = 2, \dots, l_i, \quad (66b)$$

are associated and $l_1 + \dots + l_{\rho_d} = n_d$. In (66) the vectors $\varphi_{i(k)}$ are the generalized eigenvectors of S_d w.r.t. the eigenvalue $\nu_i, i = 1, \dots, \rho_d$, in which ρ_d is the number of eigenvalues of S_d with linearly independent eigenvectors (see, e.g., Reference 23 (Chapter 6) for details on the related Jordan canonical form). The generalized eigenvectors $\varphi_{i(k)}, i = 1, \dots, \rho_d, k = 1, \dots, l_i$, are determined such that they form a basis for $\mathbb{C}^{n_d}$, which is always possible. Postmultiplying (63) by the generalized eigenvectors $\varphi_{i(k)}$, defining $r_{i(k)} = R \varphi_{i(k)}$, $g_{i(k)} = -\Lambda^{-1} \bar{G} \varphi_{i(k)}$ and $m_{i(k)} = M \varphi_{i(k)}$ lead to the one-sided coupled BVPs

$$r_{i(k)}''(z) = (\nu_i + \mu_o) \Lambda^{-1}(z) r_{i(k)}(z) + g_{i(k)}(z) + \Lambda^{-1}(z) r_{i(k-1)}(z), \quad z \in (0, 1), \quad (67a)$$

$$r_{i(k)}(0) = 0, \quad (67b)$$

$$r_{i(k)}(1) = m_{i(k)} \quad (67c)$$

and $r_{i(0)} \equiv 0$. The solution of (67a) and (67b) is

$$r_{i(k)}(z) = \Omega(\nu_i) r_{i(k)}'(0) + n_{i(k)}(z) \quad (68)$$

with $\Omega(s) = E_1^\top \Psi(1, 0, s) E_2$ and

$$n_{i(k)}(z) = \int_0^z E_1^\top \Psi(z, \zeta, \nu_i) E_2 (\Lambda^{-1}(\zeta) r_{i(k-1)}(\zeta) + g_{i(k)}(\zeta)) d\zeta. \quad (69)$$

Inserting (68) in (67c) gives

$$\Omega(\nu_i) r_{i(k)}'(0) = m_{i(k)} - n_{i(k)}(1). \quad (70)$$

Hence, the BVP (67) is solvable if

$$\det \Omega(\nu_i) \neq 0. \quad (71)$$

It is readily verified that the eigenvalues of the PDE subsystem (61b)–(61d) result from the solution of the characteristic equation $\det \Omega(s) \neq 0, s \in \mathbb{C}$, which establishes the solvability condition of (63). The pair $(R'(0), S_d)$ is observable iff

$$R'(0) \varphi_{i(1)} = r_{i(1)}'(0) \neq 0, \quad i = 1, \dots, \rho_d, \quad (72)$$

by the PBH eigenvector test (see Reference 24 (Theorem 6.2.5)). With (63) solvable, that is, (71) holds, the condition (72) is satisfied, iff the right hand side of (70) does not vanish for $k = 1$. This is the condition (86) of the lemma (see the definition of $m_{i(k)}, g_{i(k)}$ and (69)). ■

With this, the stability of the observer error dynamics (53) is clarified in the next theorem.

Theorem 2 (Disturbance observer). Suppose that $\lambda_{\max}$ is the maximum eigenvalue of the operator $Ah = \Lambda_i h''$, $i = 1, 2$, $h \in D(\mathcal{A}) = \{h \in (H^2(0, 1))^2 \mid h(0) = h(1) = 0\}$. Let $\mu_o > \lambda_{\max}$, $\bar{\mu}_1 < 0$, and furthermore let the observer gain L_d be such that $\bar{S}_d = S_d - L_d R'(0)$ is Hurwitz with $R'(0)$ determined by (63). Then, the observer error system (53) with the state $e_o(t) = \text{col}(e_d(t), e_\xi(1, t), e_\xi(t))$ and $e_\xi(t) = e_\xi(z, t)$, $z \in [0, 1]$, is exponentially stable in the norm

$$\|\cdot\|_o = (\|\cdot\|_{\mathbb{C}^{n_d}}^2 + \|\cdot\|_{\mathbb{C}^2}^2 + \|\cdot\|_{L_2^2}^2)^{\frac{1}{2}} \quad (73)$$

with the decay rate $\alpha_o = -\max(\lambda_d, \mu_1, \lambda_{\max} - \mu_o) > 0$. Hence,

$$\|e_o(t)\|_o \leq M_o e^{(-\alpha_o + c)t} \|e_o(0)\|_o, \quad t \geq 0, \quad (74)$$

holds for all $e_o(0) \in \mathbb{C}^{n_d} \oplus \mathbb{C}^2 \oplus (H^2(0, 1))^2 \subset X$ satisfying the boundary conditions, $M_o \geq 1$ and any $c > 0$.

Proof. The error system (61) is a stable ODE-PDE-ODE cascade with the decay rate $\alpha_o = -\max(\lambda_d, \mu_1, \lambda_{\max} - \mu_o) > 0$, which can be shown using the results from Reference 18. As the transformations (54) are boundedly invertible, the exponential stability of the observer error in original coordinates (53) follows readily. ■

The next section deals with the case, when both boundary agents measure their absolute and their relative position, which requires no communication (see right hand side in Figure 4). Then, two boundary observers are designed taking their respective measurement into account.

6.1.2 | Boundary observers

With the *boundary measurements*

$$\eta_0 = \begin{bmatrix} x(0) \\ x'(0) \end{bmatrix} \quad \text{and} \quad \eta_1 = \begin{bmatrix} x(1) \\ x'(1) \end{bmatrix} \quad (75)$$

the *boundary disturbance observers* for the continuum model (10) with the Hopf-Cole-type transformation (23) applied, take the form

$$\dot{\hat{v}}_{d,0} = S_d \hat{v}_{d,0} + L_{d,0}(\bar{x}'(0) - \hat{x}'(0)), \quad (76a)$$

$$\dot{\hat{x}} = D(\theta) \hat{x}'' + H(\theta) \hat{x} + \bar{u}(\theta) + \bar{G}(\theta) P_d \hat{v}_{d,0} + L_0(\theta)(\bar{x}'(0) - \hat{x}'(0)), \quad (\theta, t) \in (0, 1) \times \mathbb{R}^+, \quad (76b)$$

$$\hat{x}(0) = \bar{x}(0), \quad (76c)$$

$$\hat{x}(1) = \hat{w}_1, \quad (76d)$$

$$\dot{\hat{w}}_1 = \bar{u}(1) + \bar{G}(1) P_d \hat{v}_{d,0} + L_{1,0}(\bar{x}'(0) - \hat{x}'(0)) \quad (76e)$$

for $\theta = 0$ and

$$\dot{\hat{v}}_{d,1} = S_d \hat{v}_{d,1} + L_{d,1}(\bar{x}'(1) - \hat{x}'(1)), \quad (77a)$$

$$\dot{\hat{x}} = D(\theta) \hat{x}'' + H(\theta) \hat{x} + \bar{u}(\theta) + \bar{G}(\theta) P_d \hat{v}_{d,1} + L_1(\theta)(\bar{x}'(1) - \hat{x}'(1)), \quad (\theta, t) \in (0, 1) \times \mathbb{R}^+, \quad (77b)$$

$$\hat{x}(0) = \hat{w}_0, \quad (77c)$$

$$\hat{x}(1) = \bar{x}(1), \quad (77d)$$

$$\dot{\hat{w}}_0 = \bar{u}(0) + \bar{G}(0) P_d \hat{v}_{d,1} + L_{0,1}(\bar{x}'(1) - \hat{x}'(1)) \quad (77e)$$

for $\theta = 1$ (cf. (24)), with $\bar{x}(\theta) = T(\theta)x(\theta)$, $\bar{G}(\theta) = T(\theta)G(\theta)$ and the observer gains $L_{d,i} \in \mathbb{R}^{n_d \times 2}$ and $L_i(\theta)$, $L_{0,1}$, $L_{1,0} \in \mathbb{R}^{2 \times 2}$ to be determined in the sequel. Note, that due to the measurement (75) for $\theta = 0$ the ODE state $w_0 = \bar{x}(0)$ and for $\theta = 1$, $w_1 = \bar{x}(1)$ are known. Consequently, the corresponding boundary ODEs are not contained in the observers. The resulting observer error dynamics using $e_x = \bar{x} - \hat{x}$, $e_{w,i} = w_i - \hat{w}_i$, $i = 0, 1$ and $e_{d,i} = v_{d,i} - \hat{v}_{d,i}$ read

$$\dot{e}_{d,0} = S_d e_{d,0} - L_{d,0} e'_x(0), \quad (78a)$$

$$\dot{e}_x = D(\theta) e''_x + H(\theta) e_x + \bar{G}(\theta) P_d e_{d,0} - L_0(\theta) e'_x(0), \quad (78b)$$

$$e_x(0) = 0, \quad (78c)$$

$$e_x(1) = e_{w_1}, \quad (78d)$$

$$\dot{e}_{w_1} = \bar{G}(1) P_d e_{d,0} - L_{1,0} e'_x(0) \quad (78e)$$

for $\theta = 0$ and

$$\dot{e}_{d,1} = S_d e_{d,1} - L_{d,1} e'_x(1), \quad (79a)$$

$$\dot{e}_x = D(\theta) e''_x + H(\theta) e_x + \bar{G}(\theta) P_d e_{d,1} - L_1(\theta) e'_x(1), \quad (79b)$$

$$e_x(0) = e_{w_0}, \quad (79c)$$

$$e_x(1) = 0, \quad (79d)$$

$$\dot{e}_{w_0} = \bar{G}(0) P_d e_{d,1} - L_{0,1} e'_x(1), \quad (79e)$$

for $\theta = 1$. To stabilize (78), (79), the transformations

$$\bar{e}_{w_1} = e_{w_1} - M_0 e_{d,0} - \int_0^1 N_0(\zeta) e_x(\zeta) d\zeta, \quad (80a)$$

$$e_x = \tilde{e}_x - \int_0^\theta \bar{K}_0(\theta, \zeta) \tilde{e}_x(\zeta) d\zeta = \mathcal{T}_{0,0}^{-1}[\tilde{e}](\theta), \quad (80b)$$

$$\check{e}_x = \tilde{e}_x - R_0(\theta) e_{d,0} \quad (80c)$$

for $\theta = 0$ and

$$\bar{e}_{w_0} = e_{w_0} - M_1 e_{d,1} - \int_0^1 N_1(\zeta) e_x(\zeta) d\zeta, \quad (81a)$$

$$e_x = \tilde{e}_x - \int_\theta^1 \bar{K}_1(\theta, \zeta) \tilde{e}_x(\zeta) d\zeta = \mathcal{T}_{0,1}^{-1}[\tilde{e}](\theta), \quad (81b)$$

$$\check{e}_x = \tilde{e}_x - R_1(\theta) e_{d,1} \quad (81c)$$

for $\theta = 1$ are applied. By making use of calculations similar to Section 6.1.1, the *final target systems*

$$\dot{\bar{e}}_{w_1} = \bar{\mu}_0 \bar{e}_{w_1}, \quad (82a)$$

$$\dot{\check{e}}_x = D(\theta) \check{e}''_x - \mu_{0,0} \check{e}_x, \quad (82b)$$

$$\check{e}_x(0) = 0, \quad (82c)$$

$$\check{e}_x(1) = \bar{e}_{w_1}, \quad (82d)$$

$$\dot{e}_{d,0} = (S_d - L_{d,0} R'_0(0)) e_{d,0} - L_{d,0} \check{e}'_x(0) \quad (82e)$$

TABLE 1 The observer gains $L_i(\theta)$, $i = 0, 1$, and $L_{1,0}$, $L_{0,1}$, the kernel equations for $\bar{K}_i$ with the corresponding decoupling equations for N_i , R_i , $i = 0, 1$, on the respective domains (see Section 6.1.1).

Measurement $\eta_0 = \text{col}(x(0), x'(0))$	Measurement $\eta_1 = \text{col}(x(1), x'(1))$
$\tilde{L}_0(\theta) = R_0(\theta)L_{d,0}$	$\tilde{L}_1(\theta) = R_1(\theta)L_{d,1}$
$L_0(\theta) = \mathcal{T}_{0,0}^{-1}[\tilde{L}_0](\theta) - \bar{K}_0(\theta, 0)D(0)$	$L_1(\theta) = \mathcal{T}_{0,1}^{-1}[\tilde{L}_1](\theta) + \bar{K}_1(\theta, 1)D(1)$
$L_{1,0} = \int_0^1 N_0(\zeta)L_0(\zeta)d\zeta + M_0L_{d,0} + N_0(0)D(0)$	$L_{0,1} = \int_0^1 N_1(\zeta)L_1(\zeta)d\zeta + M_1L_{d,1} - N_1(1)D(1)$
$M_0 = (\int_0^1 N_0(\zeta)\bar{G}(\zeta)d\zeta P_d - \bar{G}(1)P_d)(\bar{\mu}_0 I - S_d)^{-1}$	$M_1 = (\int_0^1 N_1(\zeta)\bar{G}(\zeta)d\zeta P_d - \bar{G}(0)P_d)(\bar{\mu}_1 I - S_d)^{-1}$
$(N_0 D)''(\zeta) = N_0(\zeta)(\bar{\mu}_0 I - H(\zeta))$	$(N_1 D)''(\zeta) = N_1(\zeta)(\bar{\mu}_1 I - H(\zeta))$
$(N_0 D)'(1) = \bar{\mu}_0 I$	$(N_1 D)'(0) = \bar{\mu}_1 I$
$(N_0 D)(1) = 0$	$(N_1 D)(0) = 0$
$\Lambda \bar{K}_{0,\theta\theta}(\zeta) - (\bar{K}_0 \Lambda)_{\zeta\zeta}(\zeta) = (A(\zeta) - \mu_{0,0}I)\bar{K}_0$	$\Lambda \bar{K}_{1,\theta\theta}(\zeta) - (\bar{K}_1 \Lambda)_{\zeta\zeta}(\zeta) = (A(\zeta) - \mu_{0,1}I)\bar{K}_1$
$\Lambda(\bar{K}_0' + \bar{K}_{0,\theta})(\theta) + (\bar{K}_0 \Lambda)_{\zeta}(\theta) = \mu_{0,0}I + A(\theta)$	$\Lambda(\bar{K}_1' + \bar{K}_{1,\theta})(\theta) + (\bar{K}_1 \Lambda)_{\zeta}(\theta) = \mu_{0,1}I + A(\theta)$
$-N_0(\zeta) + \int_{\zeta}^1 N_0(\bar{\zeta})\bar{K}_0(\bar{\zeta}, \zeta)d\bar{\zeta} = \bar{K}_0(1, \zeta)$	$-N_1(\zeta) + \int_0^{\zeta} N_1(\bar{\zeta})\bar{K}_1(\bar{\zeta}, \zeta)d\bar{\zeta} = \bar{K}_1(0, \zeta)$
$\Lambda \bar{K}_0 - \bar{K}_0 \Lambda = 0$	$\Lambda \bar{K}_1 - \bar{K}_1 \Lambda = 0$
$D(\theta)R_0''(\theta) - \mu_{0,0}R_0(\theta) - R_0(\theta)S_d = -\tilde{G}(\theta)P_d$	$D(\theta)R_1''(\theta) - \mu_{0,1}R_1(\theta) - R_1(\theta)S_d = -\tilde{G}(\theta)P_d$
$R_0(0) = 0$	$R_1(0) = M_1$
$R_0(1) = M_0$	$R_1(1) = 0$

for $\theta = 0$ and

$$\dot{\tilde{e}}_{w_0} = \bar{\mu}_1 \tilde{e}_{w_0}, \quad (83a)$$

$$\dot{\tilde{e}}_x = D(\theta)\tilde{e}_x'' - \mu_{0,1}\tilde{e}_x, \quad (83b)$$

$$\tilde{e}_x(0) = \tilde{e}_{w_0}, \quad (83c)$$

$$\tilde{e}_x(1) = 0, \quad (83d)$$

$$\dot{e}_{d,1} = (S_d - L_{d,1}R_1'(1))e_{d,1} - L_{d,1}\tilde{e}_x'(1) \quad (83e)$$

for $\theta = 1$ are obtained. In both cases parabolic ODE-PDE-ODE cascades, with $\mu_{0,i} \in \mathbb{R}$ and $\bar{\mu}_i < 0$, $i = 0, 1$, are obtained. The resulting observer gains, following from the solution of the kernel and decoupling equations are summarized in Table 1.

The remaining observer gains $L_{d,i}$, $i = 0, 1$, are chosen such that the matrices $S_d - L_{d,i}R_i'(i)$, $i = 0, 1$, are Hurwitz to stabilize the $e_{d,i}$ -systems, $i = 0, 1$ (82e) and (83e). The next lemma presents the corresponding conditions.

Lemma 3 (Observability). *The decoupling equations for $R_i(\theta)$, $i = 0, 1$, in Table 1 have a unique C^2 -solution if any eigenvalue of S_d is not an eigenvalue of the PDE subsystems (82b)–(82d) and (83b)–(83d) respectively. Let $\Psi_i(\theta, \zeta, s) : [0, 1]^2 \times \mathbb{C} \rightarrow \mathbb{C}^{4 \times 4}$ be the fundamental matrix solving the IVP*

$$\partial_{\zeta}\Psi_i(\theta, \zeta, s) = \begin{bmatrix} 0 & I \\ (s + \mu_{0,i})D^{-1}(\theta) & 0 \end{bmatrix}\Psi_i(\theta, \zeta, s), \quad \Psi_i(\zeta, \zeta, s) = I, \quad i = 0, 1. \quad (84)$$

Then, the pair $(R_i'(i), S_d)$, $i = 0, 1$, is observable iff

$$\left(M_0 - \int_0^1 E_1^{\top}\Psi_0(1, \zeta, \nu_k)E_2D^{-1}(\zeta)\tilde{G}(\zeta)P_d d\zeta\right)\varphi_{k(1)} \neq 0, \quad k = 1, \dots, \rho_d, \quad (85)$$

for $\theta = 0$ and

$$\left(E_1^\top \Psi_1(1, \zeta, v_k) E_1 M_1 - \int_0^1 E_1^\top \Psi_1(1, \zeta, v_k) E_2 D^{-1}(\zeta) \tilde{G}(\zeta) P_d d\zeta \right) \varphi_{k(1)} \neq 0, \quad k = 1, \dots, \rho_d, \quad (86)$$

for $\theta = 1$, where $\varphi_{k(1)}$ are the linearly independent eigenvectors w.r.t. the ρ_d distinct eigenvalues v_k of S_d .

The proof of the lemma can readily be inferred from Section 6.1.1. The remaining task is to communicate the disturbance state estimates $\hat{v}_d$ and the leader states v , known to the boundary agents, to the other agents. This problem is solved in the next section.

6.2 | Cooperative signal observer

In what follows it is assumed that the boundary agents (52) know the state v of the leaders (3), as an estimate can be obtained by employing an ODE-observer using the leaders positions $r(t)$. Since the remaining agents in the network also need the states of the disturbance model (2) and the leaders (3) to realize the feedforward controller (13a), they have to be communicated from the boundary agents to each of these agents. Then, each in-domain agent i , $i = 1, \dots, N-1$, only needs to know $\Pi(\theta_i)$ specifying the agent formation. In the sequel it is sufficient to consider the case of a vanishing disturbance observer error, that is, $\hat{v}_{d,i} = v_{d,i}$, since the general case will be treated in Section 7. To ensure a scalable design, by considering the joint signal model (44) the *continuum cooperative signal observer*

$$\dot{\hat{w}}(\theta, t) = S_w \hat{w}(\theta, t) - \mu(a(\theta) \hat{w}''(\theta, t) + b(\theta) \hat{w}'(\theta, t)), \quad (\theta, t) \in (0, 1) \times \mathbb{R}^+, \quad (87a)$$

$$\hat{w}(0, t) = w(t), \quad (87b)$$

$$\hat{w}(1, t) = w(t) \quad (87c)$$

with the IC $\hat{w}(\theta, 0) = \hat{w}_0 \in \mathbb{R}$, $a, b \in C[0, 1]$, $\theta \in [0, 1]$ and the *observer gain* $\mu \in \mathbb{R}^+$ is designed. Therein, the parameter θ is a continuum agent index specifying the agent for which the cooperative observer is locally implemented after a spatial discretization (cf. Section 3). The diffusive-advective coupling in (87a) accounts for an information exchange between neighboring observers. Since the boundary agents at $\theta = 0, 1$ are informed, they have direct access to the signal model state $w(t)$ yielding the BCs (87b)–(87c). Defining the *observer error* $e_w(\theta, t) = w(t) - \hat{w}(\theta, t)$ yields the corresponding *error system*

$$\dot{e}_w = S_w e_w + \mu(a(\theta) e_w'' + b(\theta) e_w'), \quad (88a)$$

$$e_w(0) = 0, \quad (88b)$$

$$e_w(1) = 0. \quad (88c)$$

The choice of the gain μ is clarified by the next theorem.

Theorem 3 (Cooperative signal observer). *Suppose that $\lambda_{\max} < 0$ holds for the largest eigenvalue of the operator $\mathcal{A}h = ah'' + bh'$, $h \in D(\mathcal{A}) = \{h \in (H^2(0, 1))^{n_w} \mid h(0) = h(1) = 0\} \subset X$, in the state space $X = (L_2(0, 1))^{n_w}$ with the usual L_2 norm. Then, the observer error system (88) is exponentially stable in the L_2 -norm and thus satisfies $\|e_w(t)\| \leq M e^{-\alpha t} \|e_w(0)\|$, $t \geq 0$, for some $M \geq 1$ and the decay rate $\alpha = -\mu \lambda_{\max}$.*

Proof. Introduce the operator of the error system (88) $\mathcal{A}_o h = (S + \mu \mathcal{A})h$, $h \in D(\mathcal{A})$. Thus, the error system (88) can be represented by the abstract IVP $\dot{e}_w(t) = \mathcal{A}_o e_w(t)$, $t \geq 0$, $e_w(0) = e_{w,0} \in X$. By making use of Reference 25 (Lemma A.3.3) it is directly verified, that the abstract IVP is well-posed, since the operator $\mathcal{A}_o$ is Riesz spectral (see Reference 26). In order to investigate the spectrum $\sigma(\mathcal{A}_o)$, assume that the matrix $S_w \in \mathbb{R}^{n_w \times n_w}$ has the Jordan chains to which the eigenvalue problems $\omega_{i(1)}^\top S = \mu_i \omega_{i(1)}^\top$, $\omega_{i(k)}^\top S = \mu_i \omega_{i(k)}^\top + \omega_{i(k-1)}^\top$, $k = 2, \dots, l_i$, are associated and $l_1 + \dots + l_{r_w} = n_w$. Therein, the vectors $\omega_{i(k)}^\top$ are the generalized left eigenvectors of S

w.r.t. the eigenvalue μ_i , $i = 1, \dots, r_w$, in which r_w is the number of eigenvalues of S with linearly independent eigenvectors (see Reference 23 (Chapter 6)). It is assumed that the generalized left eigenvectors $\omega_{i,(k)}^\top$ form a basis in $\mathbb{C}^{n_w}$, which is always possible. Premultiplying (88a) by $\omega_{i,(k)}^\top$ and introducing the abbreviation $e_{i,(k)} = \omega_{i,(k)}^\top e_w$ yields the abstract IVP $\dot{e}_{i,(k)} = (\mu_i I + \mu \mathcal{A})e_{i,(k)} + e_{i,(k-1)}$ with $e_{i,(0)} = 0$. Since $\sigma(S) \subset j\mathbb{R}$ by assumption, the result $\max_{\lambda \in \sigma(\mu_i I + \mu \mathcal{A})} \operatorname{Re} \lambda = \mu \lambda_{\max} < 0$ follows in view of $\mu > 0$ and $\lambda_{\max} < 0$. This implies that $\mu \lambda_{\max}$ is also the largest real part of the eigenvalues for the composite system with the state $e^*(t) = \operatorname{col}(e_{1,(1)}, \dots, e_{1,(l_1)}, \dots, e_{r_w,(l_{r_w})})$. This is due to the fact that the coupling in the abstract IVP gives rise to a bounded operator so that the Riesz spectral property of $\mu_i I + \mu \mathcal{A}$ (see Reference 26) is inherited by the composite system. Consequently, the eigenvalues determine the decay rate by the spectrum determined growth assumption (see Reference 27 (Theorem 2.3.5)). The latter is related to the PDE (88a) by the invertible transformation $e^*(t) = W e_w(t)$ with $W \in \mathbb{C}^{n_w \times n_w}$, composed of the generalized left eigenvectors $\omega_{i,(k)}^\top$ of S as rows. Consequently, $\mu \lambda_{\max}$ is also the largest real part of the eigenvalues of $\mathcal{A}_o$, which proves the theorem. ■

Remark 1. Note that the condition $\lambda_{\max} < 0$ of Theorem 3 can always be achieved by a suitable choice of $a(\theta)$, $b(\theta)$ and thus is not restrictive. It is interesting to remark that $\operatorname{Re}_{\lambda \in \sigma(\mathcal{A})} \lambda < 0$ corresponds to the fact that in the finite-dimensional setup the leader-follower matrix H satisfies $\operatorname{Re}_{\lambda \in \sigma(-H)} \lambda < 0$ (see Reference 28 (Lemma 3.3)).

A spatial discretization of (87) according to Section 4 yields the *lumped cooperative signal observer*

$$\dot{\hat{w}}_i = S_w \hat{w}_i + \mu \sum_{j=0}^N a_{ij}(\hat{w}_j - \hat{w}_i), \quad i = 1, \dots, N-1. \quad (89)$$

This is a network of local signal observers (89), which are synchronized with the boundaries $\hat{w}_0 = w$ and $\hat{w}_N = w$ using a network communication. With this, the states of the disturbance model and the leader are distributed from the boundary agents to the in-domain agents.

7 | OUTPUT FEEDBACK REGULATOR

The *output feedback regulator* is obtained by using the estimates of the disturbance observers (51) and the cooperative signal observer (87) to implement the feedforward controller (13) and the state feedback controller (31), (35). The next theorem shows that this leads to a solution of the posed cooperative output regulation problem.

Theorem 4 (Stability of the closed-loop system). *Suppose the state feedback controller (31), (35) and the observers (51) or (76), (77) and (87) are designed according to Theorems 1 to 3. Then, the closed-loop system resulting from applying the output-feedback controller to the tracking-error system (27) is well-posed in the state space $X_{cl} = \mathbb{C}^2 \oplus \mathbb{C}^2 \oplus (L_2(0, 1))^2 \oplus \mathbb{C}^{n_d} \oplus \mathbb{C}^2 \oplus (L_2(0, 1))^2 \oplus (L_2(0, 1))^{n_w}$ with the usual weighted inner product and is exponentially stable in the norm $\|\cdot\|_{cl} = (\|\cdot\|_c^2 + \|\cdot\|_o^2 + \|\cdot\|_{L_2}^{n_w})^{\frac{1}{2}}$ (see (41) and (73)) with the decay rate $\alpha_{cl} = \min(\alpha_c, \alpha_o, \alpha) > 0$. In particular, the closed-loop state $e_{cl}(t) = \operatorname{col}(e_c(t), e_o(t), e_w(t))$ satisfies*

$$\|e_{cl}(t)\|_{cl} \leq M_{cl} e^{-(\alpha_{cl}+c)t} \|e_{cl}(0)\|_{cl}, \quad t \geq 0 \quad (90)$$

for all $e_{cl}(0) \in \mathbb{C}^2 \oplus \mathbb{C}^2 \oplus (H^2(0, 1))^2 \oplus \mathbb{C}^{n_d} \oplus \mathbb{C}^2 \oplus (H^2(0, 1))^2 \oplus (H^2(0, 1))^{n_w} \subset X_{cl}$, $M_{cl} \geq 1$ and any $c > 0$.

Proof. In the feedforward controller (13) $w(\theta, t)$ is replaced with its estimate $\hat{w}(\theta, t)$ from the cooperative signal observer (87) for $\theta \in (0, 1)$ and the disturbance model state $v_d(t)$ with the estimates $\hat{v}_{d,i}(t)$, $i = 1, 2$, from the disturbance observers (51) or (76), (77) for $\theta \in \{0, 1\}$ to obtain the observer-based feedforward controller

$$\ddot{u}(\theta) = M(\theta)\hat{w}, \quad \theta \in (0, 1), \quad (91a)$$

$$u(0) = \Pi(0)PSv - G(0)P_d\hat{v}_{d,1} + \ddot{u}(0), \quad (91b)$$

$$u(1) = \Pi(1)PSv - G(1)P_d\hat{v}_{d,2} + \ddot{u}(1) \quad (91c)$$

with

$$M(\theta) = \text{diag}(\Pi(\theta)PS - D(\theta)\Pi''(\theta)P - C(\theta)\Pi'(\theta)P, -G(\theta)P_d). \quad (92)$$

Use $\hat{w}(\theta, t) = w(\theta, t) - e_w(\theta, t)$, $\hat{v}_{d,i}(t) = v_{d,i}(t) - e_{d,i}(t)$ and apply the feedforward controller (91) to the continuum model (10) followed by the Hopf-Cole-type transformation (23) and the folding transformation (25) to obtain the error system

$$\dot{\varepsilon}(z, t) = \Lambda(z)\varepsilon''(z, t) + A(z)\varepsilon'(z, t) - \overline{\overline{M}}(z)\varepsilon_w(z, t), \quad \theta \in (0, 1), \quad (93a)$$

$$\varepsilon'_1(0, t) = -\tilde{\theta}_f \varepsilon'_2(0, t), \quad \varepsilon_2(0, t) = \varepsilon_1(0, t), \quad (93b)$$

$$\dot{\varepsilon}(1, t) = \begin{bmatrix} \overline{G}_0 P_d & 0 \\ 0 & \overline{G}_1 P_d \end{bmatrix} e_d(t) + v(t). \quad (93c)$$

Therein $\overline{\overline{M}}(z) = \text{diag}(\overline{M}(\theta_f - \theta_f z), \overline{M}(\theta_f + (1 - \theta_f)z))$, $\overline{M}(\theta) = T(\theta)M(\theta)$ applies and the folded cooperative signal observer error state $\varepsilon_w(z) = \text{col}(e_w(\theta_f - \theta_f z), e_w(\theta_f + (1 - \theta_f)z)) \in \mathbb{R}^{2n_w}$ and $e_d = \text{col}(e_{d,1}, e_{d,2})$ is used. For $v(t)$ use (31) and (35), replace $\varepsilon(z, t)$ with its estimate $\hat{\varepsilon}(z, t)$ and use again $\hat{\varepsilon}(z, t) = \varepsilon(z, t) - e_\varepsilon(z, t)$ followed by the transformations (28) to obtain the final target system

$$\dot{\bar{\varepsilon}}(z, t) = \Lambda(z)\bar{\varepsilon}''(z, t) - \mu_c \bar{\varepsilon}(z, t) - \bar{A}_0(z)\bar{\varepsilon}(0, t) - \bar{A}_1(z)\bar{\varepsilon}'(0, t) - \overline{\overline{M}}(z)\varepsilon_w(z, t), \quad (94a)$$

$$\bar{\varepsilon}'_1(0, t) = -\tilde{\theta}_f \bar{\varepsilon}'_2(0, t), \quad \bar{\varepsilon}_2(0, t) = \bar{\varepsilon}_1(0, t), \quad (94b)$$

$$\dot{\bar{\varepsilon}}(1, t) = \begin{bmatrix} \mu_0 & 0 \\ 0 & \mu_1 \end{bmatrix} \bar{\varepsilon}(1, t) + \begin{bmatrix} \overline{G}_0 P_d & 0 \\ 0 & \overline{G}_1 P_d \end{bmatrix} e_d(t) - \begin{bmatrix} E_1^\top \mathcal{K}_1 [e_\varepsilon(z, t), e_\varepsilon(1, t)] \\ E_2^\top \mathcal{K}_1 [e_\varepsilon(z, t), e_\varepsilon(1, t)] + \mathcal{K}_2 [e_\varepsilon(z, t), e_\varepsilon(1, t)] \end{bmatrix} \quad (94c)$$

with the operators $\mathcal{K}_i$, $i = 1, 2$ from (31) and (35). By considering the transformations (54) one can replace $e_\varepsilon(z, t) = \mathcal{T}_0^{-1}[\tilde{e}_\varepsilon(\cdot, t)](z)$, $\tilde{e}_\varepsilon(z, t) = \check{e}_\varepsilon(z, t) + R(z)e_d(t)$ and $e_\varepsilon(1, t) = \bar{e}_\varepsilon(1, t) + Me_d(t) + \int_0^1 N(\zeta)e_\varepsilon(\zeta, t)d\zeta$. Therefore if Theorems 1 to 3 are fulfilled the exponentially stable systems (61) and (88) couple into (94) this results in a cascade of exponentially stable systems. Hence, using Reference 18 the closed-loop system can be represented by the abstract IVP $\dot{e}_{cl}(t) = \mathcal{A}_{cl}e_{cl}(t)$, $t > 0$, $e_{cl}(0) = D(\mathcal{A}_{cl}) \subset X_{cl}$. By making use of Reference 18 (Theorem 3), it can be verified that $\mathcal{A}_{cl}$ is the generator of an exponentially stable and analytic C_0 -semigroup with the growth rate α_{cl} . Finally, the boundedly invertible transformations lead to (90). ■

In Theorem 4 only the convergence in the L_2 -norm is proven. Using the results in Reference 29 point-wise convergence can also be shown, which fulfils (10).

Hence, using the estimates from the disturbance observer (51) or (76) and the estimates from the discretized cooperative signal model observer (87) in (45) yields the *finite-dimensional networked controller* (7) for (1) (see Figure 1), which solves the containment control problem (5). Therein $(m_i^v)^\top$ and $(\pi_i^v)^\top$ follow from spatial discretization of (91) and $\hat{x}_i^v$, $\bar{w}_i = \text{col}(v, \hat{v}_d)$ are obtained from the disturbance observer from Section 6 and the knowledge of the leader states v . The error resulting from discretizing the continuum model converges with $\mathcal{O}(1/N^2)$, when a 3-point-centered finite difference method is being used (see Reference 30 (Appendix A)). With the same rate converges the error resulting from discretizing the integral terms in the control laws (31) and (35) using the trapezoidal rule (see Reference 31). However, N depends in this case on the discretization of the infinite-dimensional disturbance observer, which can be chosen arbitrarily.

Remark 2. It is interesting to note, that the main advantage using the continuum model is the scalable design of the disturbance observer (51) or (76). With this, all agent positions are available to the boundary agents allowing problems to impose any desired synchronization dynamics via a state feedback of all agent positions. This is not possible without a continuum model, as this would result in a non-scalable design.

8 | EXAMPLE

8.1 | Problem formulation

To illustrate the results of the article, a MAS consisting of nine agents, that is, $N = 8$, with the disturbance input vector $G_i = [g_1(i/N), g_2(i/N)]^\top$, $i = 0, \dots, 8$, $g_1(z) = 0.5 - (z - 0.5)^2$ and $g_2(z) = 0.25 - 2z$, is considered. The continuum model (10) is obtained with the diffusive-advective coupling

$$D(\theta) = \begin{bmatrix} 1 + \theta & 0 \\ 0 & \frac{3}{4} - \frac{\theta}{4} \end{bmatrix} \quad \text{and} \quad C(\theta) = \begin{bmatrix} 2 & 0 \\ 0 & 4.3 \end{bmatrix} \quad (95)$$

of the agents. The disturbance $d(t)$ is constant, which leads to the disturbance model $\dot{v}_d = 0$, $v_d(0) \in \mathbb{R}$. The containment area is spanned by $\ell = 4$ leaders, which encircle the origin with an angular speed ω . This can be described with the leader model

$$\dot{v}_i = \begin{bmatrix} 0 & \omega \\ -\omega & 0 \end{bmatrix} v_i, \quad v_i(0) \in \mathbb{R}^2, \quad (96a)$$

$$r_i = v_i \quad (96b)$$

for $i = 1, \dots, 4$ with $\omega = 2/3$. The desired formation of the continuum model (10) to solve the containment control problem is given by the Bézier curve $\Pi(\theta)r(t)$ and is described by 4 control points $B_k = \alpha_k^\top \otimes I_2$, $k = 0, \dots, 3$ with $\alpha_0 = [0.75, 0, 0.25, 0]$, $\alpha_1 = [0.4, 0, 0.1, 0.5]$, $\alpha_2 = [0.05, 0, 0.45, 0.5]$, $\alpha_3 = [0.25, 0, 0.75, 0]$. This completely determines the feedforward controller (13).

8.2 | Tracking controller

Due to the choice of $D(\theta)$ and $C(\theta)$ in the continuum model, the tracking error has the maximum eigenvalue $\lambda_{\max} = 0.52$, which necessitates the design of a stabilizing boundary controller. The stability parameters in the controller design are $\mu_c = 1.5$, $\mu_0 = -3$, $\mu_1 = -2$ and the folding point is set to $\theta_f = 0.5$. The solution of the kernel equations for the kernels $K(z, \zeta)$, $Q(z, \zeta)$ and $P(z, \zeta)$ are obtained by a successive approximation with a discretization of 31 points per dimension. The boundary value problem for $\Gamma(z)$, resulting from the decoupling transformation (28b), is solved with 51 points using the MATLABs ODE-solver `ode15s`.

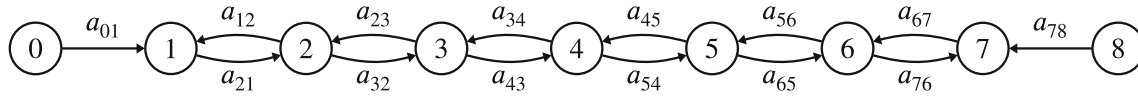
8.3 | Observer design

To reject the disturbance using the feedforward controller and to realize the tracking controller, estimates of the disturbance state $v_d(t)$ and the continuum agent state $x(\theta, t)$ are needed. Therefore, a folding disturbance observer (51) is designed. With the measurement taken at $\theta_m = \theta_f = 0.5$, this yields two disturbance observers estimating the state $x(\theta)$ in $[0, 0.5]$ and $[0.5, 1]$. The design parameters are set to $\mu_{1,i} = -9$, $\mu_{0,i} = -4$, $i = 1, 2$, and the eigenvalues of $S_d - L_{d,i}R'_i(0)$, $i = 1, 2$, are -10 for both observers. The resulting kernel equations for $\bar{K}(z, \zeta)$ are solved by successive approximation with 31 points and the boundary value problems for $N(z)$ and $R(z)$ are solved using `ode15s` with 51 points.

For the application of the feedforward controller to the in-domain agents, the signal model state $w(t)$, which is known by the boundary agents, has to be communicated to the remaining five agents. To this end, the cooperative continuum signal observer (87) with the parameters $a(\theta) = b(\theta) = 1$ is designed. This choice of cooperative coupling leads to a maximum eigenvalue of the corresponding operator $\mathcal{A}$ at $\lambda_{\max} = -8.9$ so that $\mu = 1$ suffices to stabilize the observer error dynamics.

8.4 | Finite-dimensional networked controller

The finite-dimensional network controller results from a three-point-centralized finite difference method leading to a communication topology with the line graph



where the edge weights are obtained by discretizing the diffusive-advective coupling introduced in (10). Therein, agents 0 and 8 are the controlled boundary agents, that are roots of the graph. In this case a three-point-centered finite difference scheme was used. Note, that the edge weights differ for the communication in the x -direction compared to the y -direction which results from the different diagonal entries in $D(\theta)$ and $C(\theta)$ in (95).

8.5 | Simulation results

For the simulation, the ICs of the agents, the disturbance state and the leader state are set to $x_i(0) = [1 - 0.2i, 1]^T$, $i = 0, \dots, 8$, $v_d(0) = 1$, $v_1(0) = [-0.75, -0.75]^T$, $v_2(0) = [-1.5, 0]^T$, $v_3(0) = [-0.75, 0.75]^T$ and $v_4(0) = [0.05, 0.05]^T$. The ICs of the cooperative signal observer and the disturbance observers are all set to zero. The simulation is performed with a temporal discretization of 51 points per time unit. Figure 5 shows the simulation of the MAS in the plane in 6 snapshots. In the beginning all agents are located on a line outside the containment area. From there the MAS is converging with the desired speed towards the Bézier curve, although a disturbance is acting. At the 5th snapshot all agents have reached

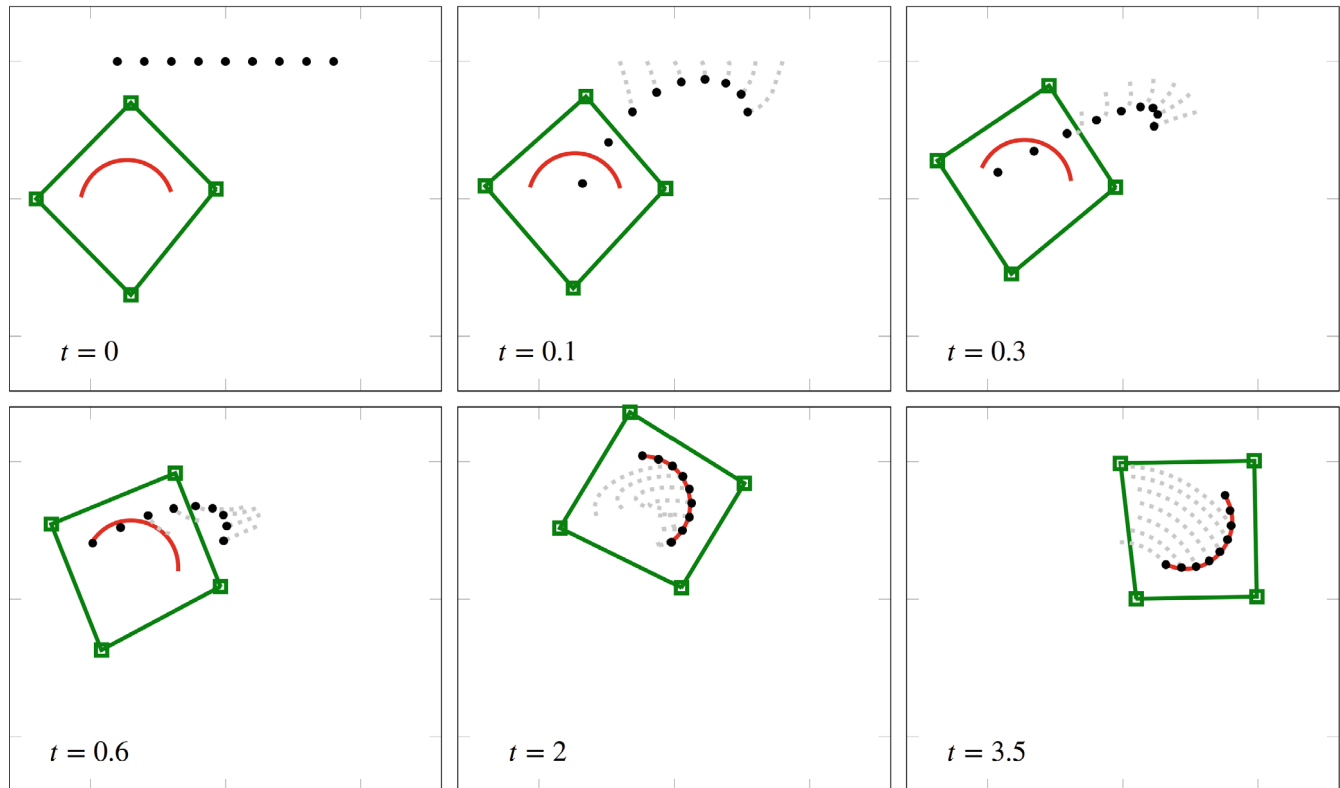


FIGURE 5 Temporal evolution of the MAS shown in six snapshots with the track (gray dotted) in between. Starting from the IC at $t = 0$, where all agents (black dots) are located on a line outside the containment area (green), the MAS is approaching the Bézier curve (red) and therefore the desired formation, in the following snapshots. From the 5th snapshot the MAS remains on the Bézier curve, which solves the containment control problem.

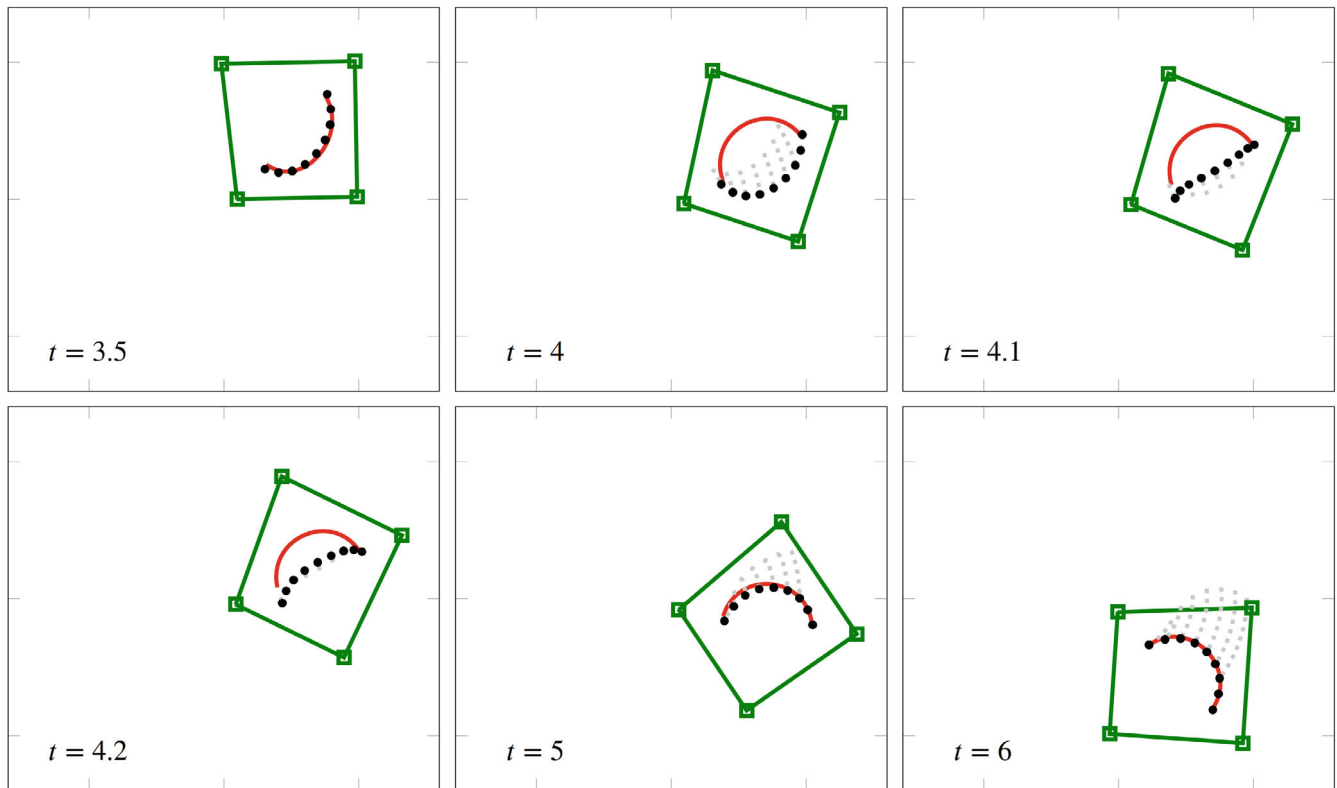


FIGURE 6 Temporal evolution of the MAS with the track (gray dotted) in between. Starting without tracking error at $t = 3.5$, where all agents (black dots) are located on their reference point on the Bézier curve (red). At $t = 4$, the Bézier curve changes and therefore the desired formation. In the following snapshots the MAS returns to its reference and remains on the Bézier curve thereafter.

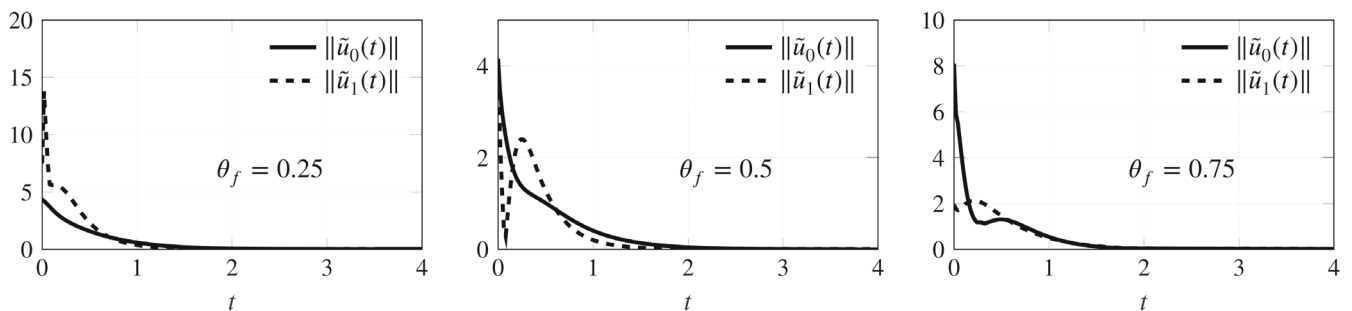


FIGURE 7 Euclidean norm of the control laws $\tilde{u}_0(t)$ and $\tilde{u}_1(t)$ to stabilize the error system (12) for different folding points $\theta_f \in \{0.25, 0.5, 0.75\}$. The solid lines show the control effort at $z = 0$ and the dashed ones at $z = 1$.

their specified position on the Bézier curve and remain there for the rest of the simulation. This solves the containment problem. Figure 6 depicts the behavior of the MAS when the Bézier curve changes. Note, that this has to be known by each agent a priori, that is, it is not communicated online. Obviously, the resulting tracking error quickly decays, which demonstrates that online formation changes are no problem for this setup, as only the new control points have to be communicated to the agents.

In Figure 7 the impact on the control effort depending on the choice of the folding point θ_f is investigated. To this end, the Euclidean norm of the control laws $\tilde{u}_i(t)$, $i = 0, 1$, which stabilize the error system (12), for the folding points $\theta_f = \theta_m \in \{0.25, 0.5, 0.75\}$ are plotted. It is clearly visible that the closer the folding point is located at a boundary agent, the less control effort this agent has to apply. At this point it is emphasized that the control effort for $\theta_f = 0.5$ is not distributed equally between the two boundary agents. The reason is the spatial dependency of the parameters $D(\theta)$ and $C(\theta)$.

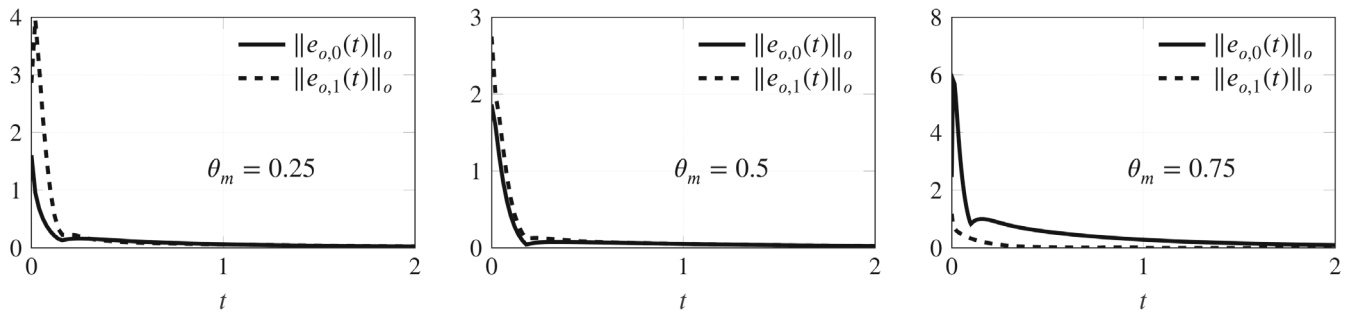


FIGURE 8 The norm (73) of the observer errors of the disturbance observers designed using the folding approach for the measurement points $\theta_m = \{0.25, 0.5, 0.75\}$. Therein $\|e_{o,1}(t)\|_o$ means the observer error of the observer at $z = 0$ and $\|e_{o,2}(t)\|_o$ is the observer error at $z = 1$.

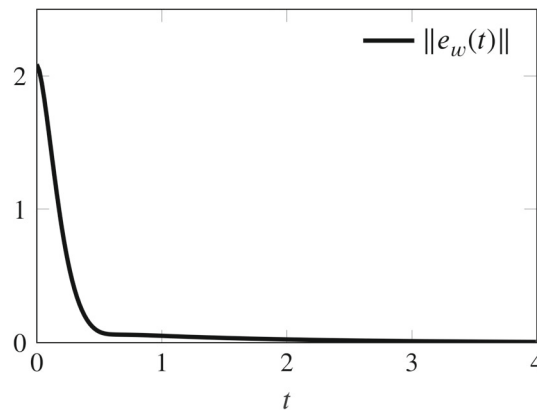


FIGURE 9 L_2 -norm of the observer error (88) of the continuum cooperative signal observer.

To investigate the impact of the measurement point θ_m on the observer error, the norm (73) of the observer error $\|e_o\|_o = (\|e_\xi(1)\|_{\mathbb{C}^2}^2 + \|e_d\|_{\mathbb{C}^{n_d}}^2 + \|e_\xi\|_{L_2}^2)^{\frac{1}{2}}$ for the measurement points $\theta_m = \{0.25, 0.5, 0.75\}$ is shown in Figure 8. The reason the decay of the shown observer errors differ depending on the measurement point θ_m is due to the difference in the folded parameters, which then result in different solutions for the kernel equations 60 and the boundary value problems 56 and 63. Finally, the L_2 -norm of the observer error (88) of the continuum cooperative signal observer for the measurement point $\theta_m = 0.5$ is shown in Figure 9. Clearly, the estimates of the signal model states w used for the feedforward quickly approach the desired values.

9 | CONCLUDING REMARKS

Future research work concerns the application of switching communication topologies to the proposed control method. Also, agents with double-integrator dynamics, which lead to a wave equation as a hyperbolic continuum model can be investigated. Furthermore, the desired formation could be extended to Bézier surfaces resulting in a 2D-formation moving in 3D-space. For this setup, backstepping results for PDEs on higher-dimensional spatial domains can be applied.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

ORCID

Joachim Deutscher  <https://orcid.org/0000-0001-6596-6066>

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